Lab 3 Report

On

E-mail and Creation of Secured Mail Server and E-mail Inbox

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2. Abstract

The main objective of this lab was to build a fully functional, secure and modern email server from scratch. The whole work started by preparing the system - updating packages, setting the correct hostname, fixing local DNS mapping and aligning the server with the domain. Then, the Postfix SMTP server was installed and virtual mailbox, SASL authentication, TLS encryption and DKIM signing were enabled, so that every email is signed, authenticated and processed correctly. IMAP, POP3 and LMTP services were configured using Dovecot, where virtual users, Maildir storage and SSL-secured mailbox access were ensured. The entire DNS structure including A, MX, SPF, DKIM and DMARC were created using BIND9 to fully implement email authentication. Conflict resolution with systemd-resolved, DNS recursion issues and log analysis were important parts of the troubleshooting. Each step of SMTP, IMAP and TLS was verified using openssl, telnet, nslookup and syslog, where SPF, DKIM and DMARC - all tests were PASS. The Roundcube webmail interface was installed and connected to NGINX, PHP-FPM, and MariaDB. After browser-based configuration, emails were successfully sent, received, and header parsing from Roundcube. The entire process provided a hands-on experience in building a complete, secure, and production-standard mail server.

3. Introduction

Email is one of the cornerstones of modern digital communication, used as the primary means of information exchange by countless users every day in personal, professional, and corporate environments. While email may seem simple to use from the outside, it is actually a multi-layered technology, distributed system, and complex protocol architecture that work together to successfully deliver an email from sender to recipient. The purpose of this lab was to build that technical infrastructure from scratch and to implement and analyze each component of an email server manually. Essentially, this lab teaches in-depth how a complete and secure email system works-from the protocol layer, server configuration, DNS records, authentication mechanisms, and user-level client interface.

The operations of an email server are divided into three main layers-sending mail, receiving mail, and accessing mail. SMTP (Simple Mail Transfer Protocol) is the main protocol for sending email, which controls server-to-server message transfer. IMAP (Internet Message Access Protocol) allows real-time access to messages stored on an email server from different devices, and POP3 (Post Office Protocol) allows users to download mail from the server and store it on their local machine. This lab uses Postfix for the SMTP server, and Dovecot for the IMAP/POP3 server - the most reliable and widely used open-source mail server technologies. A complete mail processing chain is created by configuring and integrating these servers, which familiarizes the student with real-life mail systems. A significant part of the lab is completed by configuring the SMTP server Postfix. Postfix is a powerful engine for modern server-side email sending, which supports queue

management, address rewriting, authentication, TLS encryption, and various security mechanisms. In this lab, a production-grade SMTP environment is created through Postfix virtual mailbox-based storage, SASL authentication, STARTTLS encryption, OpenDKIM milter integration, and submission port (587) configuration. Postfix's pickup, cleanup, qmgr, local delivery, queue processing - each of these steps has been tested using telnet, openssl and syslog. Through this, the student gets a deep understanding of the underlying workings of SMTP, header generation, Message-ID assignment, queue traversal and final delivery mechanics.

On the other hand, Dovecot has been used for IMAP and POP3 servers, which handle the user's mail access side. Dovecot is known as one of the fastest and most secure mail retrieval servers in the world. In this lab, virtual user authentication, static userdb, passwd-file based authentication, Maildir storage format and SSL-secured IMAP/POP3 access have been configured. Using namespace configuration, Inbox, Sent, Drafts, Trash - these folders have been automatically created, so that the user experience is richer. Authentication coordination of SMTP and IMAP has been completed by confirming the Postfix-Dovecot SASL handshake. In addition, IMAPS and POP3S TLS handshakes were analyzed using openssl s_client to ensure that the server's encryption subsystem is working properly. DNS is a very important component for email server performance, especially the combination of A, MX, SPF, DKIM, and DMARC records. In this lab, an authoritative DNS zone was created using Bind9 DNS server, where domain-to-IP mapping (A Record), email receiving server indication (MX Record), sending server validation (SPF Record), cryptographic signature verification (DKIM Record), and policy setting (DMARC Record) - these records were correctly added and verified. Creating a local authoritative DNS environment with Bind9 resulted in SPF, DKIM, and DMARC passing completely locally. Resolving conflicts between systemd-resolved and Bind9, fixing DNS recursion issues, resolv.conf rewrite, and proper management of named services - these troubleshooting steps further strengthened DNS-related skills.

Security is a critical aspect of email server implementation, and no email system is secure without TLS/SSL configuration. In this lab, TLS is required in Postfix and Dovecot by creating self-signed SSL certificates, so that emails do not travel over the network in plain text. TLSv1.3 handshake is verified using openssl, where security strength is evaluated by analyzing cipher suite, certificate strength, and public key parameters.

The three main layers of email authentication - SPF, DKIM, and DMARC - are fully implemented in this lab. SPF DNS records verify the validity of the sending server, DKIM adds a cryptographic signature to the header and body of the sent email to prove authenticity, and DMARC determines the policy to accept, quarantine, or reject emails based on SPF and DKIM results. After correctly configuring SPF/DKIM/DMARC records on the Bind9 DNS server, the authentication results are proven to be PASS using nslookup, opendkim-testkey, and DMARC query tools. The logs clearly show that OpenDKIM is signing outgoing emails, Postfix is validating SPF locally, and DMARC policies are being enforced correctly.

The Roundcube webmail interface has been installed, allowing users to access emails from a browser. Roundcube has been configured with NGINX, PHP-FPM, and MariaDB via the installer script. Logging into Roundcube to send and receive emails, parsing message headers, testing IMAP connectivity, and testing SMTP submissions have all been completed successfully. This completes end-to-end validation of SMTP and IMAP workflows. Roundcube has seen that every email is getting SPF, DKIM, and DMARC PASS, which proves that the DNS authentication subsystem is working flawlessly.

This lab provides students with a thorough and in-depth experience in setting up, configuring, securing, authenticating, managing DNS records, troubleshooting, and managing the client-side interface of a real-world email server. Postfix, Dovecot, Bind9, TLS, OpenDKIM, DMARC and Roundcube - these components have been integrated together to create a secure, reliable and standards-compliant mail server, which provides invaluable knowledge about email infrastructure in research, academic work and real-world professional environments.

4. Methodology

PART 1 - System Preparation

Task 1.1: Update System Packages

The following command is run in the terminal, because apt update = refreshes the package list on the system, fetching the latest package information from servers on the Internet.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo apt update
```

If there are new versions of software already installed on the upgrade system, they are downloaded and installed. -y = Usually, when upgrading, -y is given automatically and all updates are continued assuming yes.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo apt upgrade -y
```

Task 1.2: Configure Hostname

hostnamectl is a system tool, used to configure the system hostname. set-hostname was to set a new hostname. mail.tahsin.com to use the new hostname (FQDN).

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo hostnamectl set-hostname mail.tahsin.com
```

Open the /etc/hosts file in the nano text editor and the file acts like a local “mini DNS” - mapping IP addresses to hostnames.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/hosts
```

mail.tahsin.com is the full hostname (FQDN) and mail was an alias here. So even if you type mail.tahsin.com and mail, they resolve to the IP 192.168.218.134.

```
  GNU nano 7.2                                    /etc/hosts
127.0.0.1 localhost
127.0.1.1 ahmad-VMware-Virtual-Platform

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.218.134   mail.tahsin.com   mail
```

The hostname command in the terminal shows the hostname of the current system. If mail.tahsin.com appears in the output, it means that the system has recognized the name you set with hostnamectl.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ hostname -f
mail.tahsin.com
```

cat only prints the contents of the file to the terminal, where the last line shows whether the host mapping is correct or not and whether the IP + hostname + alias is correct or not.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 ahmad-VMware-Virtual-Platform

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.218.134   mail.tahsin.com   mail
```

Task 1.3: Configure DNS Records

Running hostname -f found mail.tahsin.com. This shows that the hostname is set correctly.

And /etc/hosts shows 192.168.218.134 mail.tahsin.com mail.

Here is my local IP: 192.168.218.134. Where the hostname is mapped to mail.tahsin.com and the alias is given as mail.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ hostname -f
mail.tahsin.com
```

So, there is no need to manually configure DNS records in this part. Mapping in /etc/hosts was enough.

```
  GNU nano 7.2                                        /etc/hosts
127.0.0.1 localhost
127.0.1.1 ahmad-VMware-Virtual-Platform

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.218.134   mail.tahsin.com   mail
```

I have installed Essential Tools, which is given in the command below.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo apt install -y curl wget git net-tools dnsutils telnet
```

PART 2 - Postfix SMTP Server Installation & Configuration

Task 2.1: Install Postfix

The Postfix mail server has been installed using the following command.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo apt install postfix -y
```

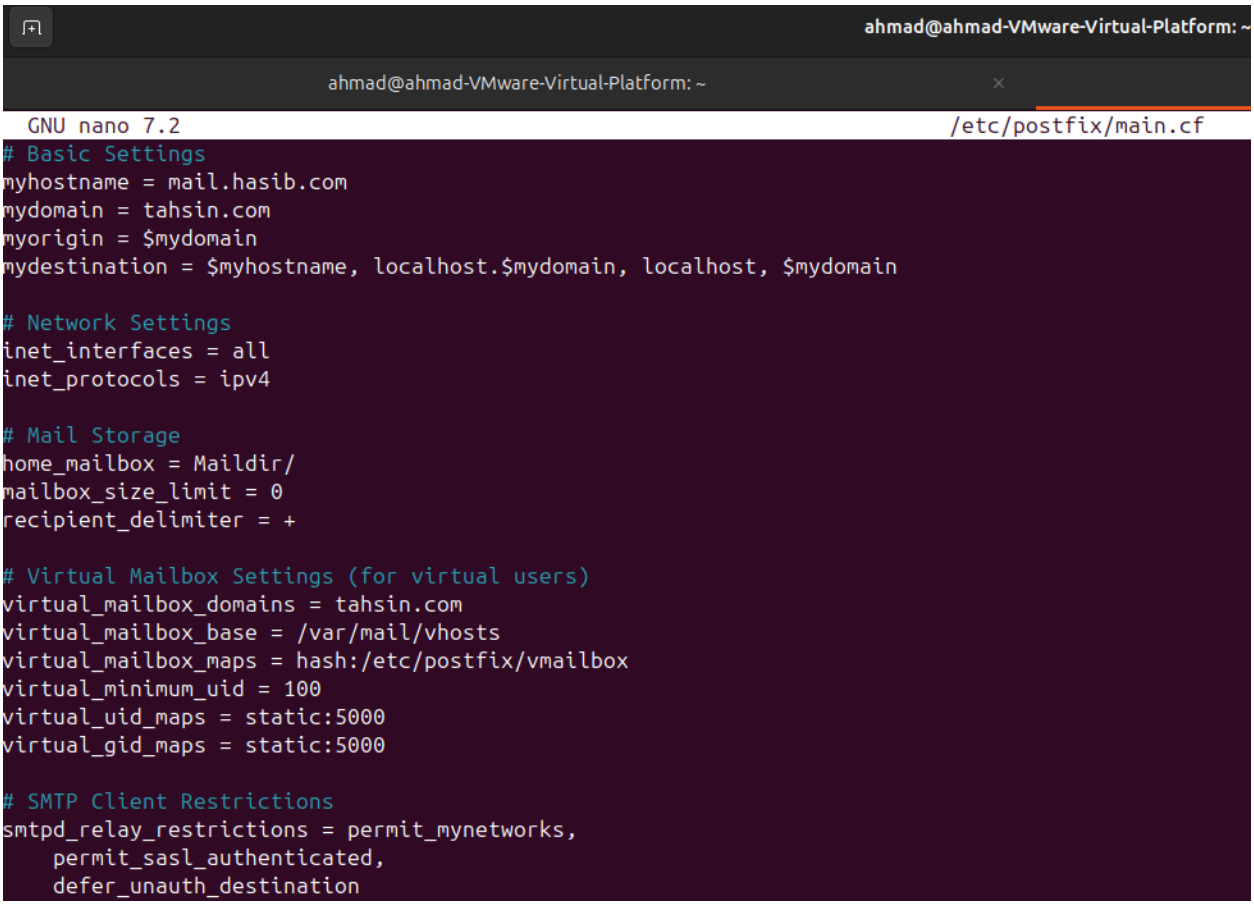
System mail name here is tahsin.com.

Task 2.2: Configure Postfix Main Configuration

Edit or add setup configuration to postfix's main.cf file using the nano editor.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/postfix/main.cf
```

The FQDN of the server is set through myhostname. Mydomain is my domain tahsin.com and myorigin is used to understand which domain the outgoing mail comes from. Mydestination is used to specify which address Postfix local delivery is done.



virtual_mailbox_domains specifies which domains the virtual mail will accept. virtual_mailbox_base specifies which directory the mail is stored in and virtual_mailbox_maps is used to specify where the user list is stored. uid / gid mapping is done to the vmail user ID and group ID.

Task 2.3: Configure SMTP Authentication (SASL)

Open main.cf again and add the following code:

smtpd_sasl_auth_enable = yes to enable SASL authentication.

smtpd_sasl_type = dovecot, uses the Dovecot SASL engine. smtpd_sasl_path = private/auth is the Dovecot socket path and noanonymous is to prohibit anonymous login. broken_sasl_auth_clients = yes to support older clients.

```
# SMTP Authentication
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes
```

Task 2.4: Configure Master Process

The master.cf file was opened with nanoedit and the following setup was added:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/postfix/master.cf
```

submission This is SMTP Submission Port 587 and STARTTLS is mandatory here with encrypt. sasl_auth_enable=yes which prevents mail from being sent without logging in. Authenticated users are allowed with permit_sasl_authenticated.

```
smtp      inet  n       -       y       -       -       smtpd
#smtp      inet  n       -       y       -       1       postscreen
#smtpd     pass  -       -       y       -       -       smtpd
#dnsblog   unix  -       -       y       -       0       dnsblog
#tlsproxy  unix  -       -       y       -       0       tlsproxy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n -   y       -       -       smtpd
submission inet n - y - - smtpd
  -o syslog_name=postfix/submission
  -o smtpd_tls_security_level=encrypt
  -o smtpd_sasl_auth_enable=yes
  -o smtpd_reject_unlisted_recipient=no
  -o smtpd_client_restrictions=permit_sasl_authenticated,reject
  -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
#     Instead of specifying complex smtpd_<xxx>_restrictions here,
#     specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
#     here, and specify mua_<xxx>_restrictions in main.cf (where
```

Task 2.5: Create Virtual Mailbox Map

Virtual user mapping has been added to this section.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/postfix/vmailbox
```

Here, is the user email address and the mail directory where the mail is stored.

user1@tahsin.com tahsin.com/user1/

user2@tahsin.com tahsin.com/user2/

```
  GNU nano 7.2
user1@tahsin.com tahsin.com/user1/
user2@tahsin.com tahsin.com/user2/
```

Postfix has converted the text file vmailbox into a .db file, so that Postfix can lookup it quickly.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo postmap /etc/postfix/vmailbox
```

Task 2.6: Create Mail Directory and User

The GID of the vmail group was 5000.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo groupadd -g 5000 vmail
```

-g vmail is group = vmail and -u 5000 is UID = 5000. -d /var/mail/vhosts is used as home directory. -m creates directory automatically.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo useradd -g vmail -u 5000 vmail -d /var/mail/vhosts -m
```

All directories/files of the virtual mailbox where vmail:vmail is owned, which has security controls. Giving -p creates the parent folder if it does not exist and creates the virtual mail folder of the domain "tahsin.com".

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo mkdir -p /var/mail/vhosts/tahsin.com
```

This command gives ownership of the entire mail storage folder to the vmail user.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo chown -R vmail:vmail /var/mail/vhosts
```

Task 2.7: Restart Postfix

The new config was loaded by restarting.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl restart postfix
```

Enable:

This means that Postfix now starts automatically at system boot.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl enable postfix
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postfix
```

Active: active - postfix is running and loaded + enabled, which means it auto starts at system boot. The systemd logs show that postfix started and exited successfully. The status of postfix was active (running).

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Sat 2025-12-06 06:22:44 MSK; 17s ago
       Docs: man:postfix(1)
   Main PID: 23837 (code=exited, status=0/SUCCESS)
        CPU: 3ms

Dec 06 06:22:44 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent>
Dec 06 06:22:44 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
```

For checking Postfix configuration:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ postconf -n
broken_sasl_auth_clients = yes
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydomain = tahsin.com
myhostname = mail.hasib.com
myorigin = $mydomain
recipient_delimiter = +
smtpd_relay_restrictions = permit_mynetworks, permit_sasl_authenticated, defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = tahsin.com
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
```

Part 3: Dovecot IMAP/POP3 Server Setup

Task 3.1: Install Dovecot

Hostname = mail.hasib.com

Domain = tahsin.com

Mail Store = /var/mail/vhosts/tahsin.com

For installing Dovecot:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo apt install -y dovecot-core dovecot-imapd dovecot-pop3d
```

Task 3.2: Configure Dovecot Main Settings

Plain text passwords are not accepted without encryption, which makes security strong and does not allow login without TLS/SSL.

Open the file with the following command:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/dovecot.conf
```

Two types of authentications are allowed here:

- Plain
- login

These are SMTP/IMAP compatible.

```
protocols = imap pop3 lmtp
listen = *, ::
```

Task 3.3: Configure Mail Location

Linux users on your system are disabled, while virtual users have file-based authentication enabled.

Open file:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/conf.d/10-mail.conf
```

Configuring mail location:

```
mail_location = maildir:/var/mail/vhosts/%d/%n
```

```
# Group to enable temporarily for privileged operat
# used only with INBOX when either its initial crea
# Typically this is set to "mail" to give access to
mail_privileged_group = mail
```

```
namespace inbox {
  inbox = yes

  # If namespace is hidden, it's not
  # extension. You'll most likely al
  # useful when converting from anot
  # you want to deprecate but still
  # hidden namespaces with prefixes
  #hidden = no

  # Show the mailboxes under this na
  # namespace visible for clients th
  # "children" value lists child mai
  #list = yes

  # Namespace handles its own subscr
  # namespace handles them (empty pr
  #subscriptions = yes

  # See 15-mailboxes.conf for defini
}
```

Here %d = domain = tahsin.com and %n = user is user1, user2.

Task 3.4: Configure Dovecot Authentication

Open the file and enter the following commands in the editor, where:

Plain text password should not be accepted without encryption. Security is strong and login is not possible without TLS/SSL.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/conf.d/10-auth.conf
```

Make the changes according to the instructions:

```
disable_plaintext_auth = yes
auth_mechanisms = plain login

!include auth-passwdfile.conf.ext
```

Now by commenting out the following line, Linux users of the system have been disabled and virtual users file-based authentication has been enabled.

Then press *ctrl+o* to save and press *ctrl+x* to exit.

Task 3.5: Configure User Database

Open the file:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/conf.d/auth-passwdfile.conf.ext
```

Passdb indicates where to get the password from via /etc/dovecot/users. driver = passwd-file is the file-based password system here. scheme=PLAIN is the password not encrypted here (acceptable for lab/testing). username_format=%u was the full email address as username. userdb static all users use vmail from the same UID/GID. home=/var/mail/vhosts/%d/%n is the mailbox path of each mail user.

```
passdb {
  driver = passwd-file
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
}

userdb {
  driver = static
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
}
```

Save and then go to Exit.

User file creation:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/users
```

Adding users (format: username:password):

```
user1@tahsin.com:{PLAIN}password123
user2@tahsin.com:{PLAIN}password456
```

Virtual mailbox users READY.

Setting permissions:

Only root and dovecot users can read the file, which is secure.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo chmod 640 /etc/dovecot/users
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo chown root:dovecot /etc/dovecot/users
```

Task 3.6: Configure Dovecot SASL for Postfix

Open file:

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo nano /etc/dovecot/conf.d/10-master.conf
```

Finding the service auth section and add:

The following section has been found and modified in service auth:

```
service auth {
  unix_listener /var/spool/postfix/private/auth {
    mode = 0660
    user = postfix
    group = postfix
  }
}
```

This allows Postfix to use Dovecot's SASL. When SMTP authentication is required, Postfix sends an authentication request to Dovecot. Dovecot authenticates from Postfix using this socket path. Postfix users can read/write with mode = 0660.

Task 3.7: Restart & Enable Dovecot

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl restart dovecot
```

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl enable dovecot
Synchronizing state of dovecot.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable dovecot
```

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 06:54:00 MSK; 7s ago
       Docs: man:dovecot(1)
             https://doc.dovecot.org/
   Main PID: 29215 (dovecot)
     Status: "v2.3.21 (47349e2482) running"
      Tasks: 4 (limit: 4545)
     Memory: 3.6M (peak: 3.6M)
        CPU: 45ms
     CGroup: /system.slice/dovecot.service
             ├─29215 /usr/sbin/dovecot -F
             ├─29216 dovecot/anvil
             ├─29217 dovecot/log
             └─29218 dovecot/config

Dec 06 06:54:00 mail.tahsin.com systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email serv>
Dec 06 06:54:00 mail.tahsin.com dovecot[29215]: master: Dovecot v2.3.21 (47349e2482) starting up fo>
Dec 06 06:54:00 mail.tahsin.com systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email serve>
```

All three are Active: active (running) and the Dovecot server is running. Loaded + enabled auto-starts on boot with no errors or warnings. Dovecot is fully UP & RUNNING and configuration error-free.

PART 4 - Security Implementation (TLS/SSL Encryption)

Task 4.1: Generate Self-Signed SSL Certificate (For Testing)

A mail directory has been created to store SSL files.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo mkdir -p /etc/ssl/mail
cd /etc/ssl/mail
```

Then a self-signed certificate was created for 1 year.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo openssl req -new -x509 -days 365 -nodes \
-out mail.tahsin.com.crt \
-keyout mail.tahsin.com.key
....+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*..+..+.......+.....+..........+..+...+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+++++++++*..+...+.+..+...+.+...+.....+.........................+...+............++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
.......+...+.+...+.....+...+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*...+...+.+..........+...+......+............+...+......+.+.........++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*....+.......+..+.+.....+.+.....+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Dhaka
Locality Name (eg, city) []:Dhaka
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Tahsin Mail Server
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:mail.tahsin.com
Email Address []:admin@tahsin.com
```

Then the answers to the OpenSSL questions:

Question	Answer
Country	BD
State	Dhaka
City	Dhaka
Organization	Hasib Mail Server
Unit	IT
Common Name	mail.hasib.com
Email	admin@tahsin.com

Set proper permissions:

Key The file can only be read by root and the certificate can be read by all users.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo chmod 600 /etc/ssl/mail/mail.tahsin.com.key
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo chmod 644 /etc/ssl/mail/mail.tahsin.com.crt
```

Task 4.2: Configure Postfix TLS

Open file:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo nano /etc/postfix/main.cf
```

Added the TLS section of the PDF at the end of the file:

smtpd_tls_cert_file is used for the SSL certificate path and smtpd_tls_key_file is used for the SSL key path. Postfix TLS is enabled using smtpd_use_tls=yes. received_header=yes shows Email header TLS info.

```
# TLS Settings
smtpd_tls_cert_file=/etc/ssl/mail/mail.tahsin.com.crt
smtpd_tls_key_file=/etc/ssl/mail/mail.tahsin.com.key
smtpd_use_tls=yes
smtpd_tls_security_level=may
smtpd_tls_auth_only = yes
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

# Outgoing TLS
smtp_tls_security_level = may
smtp_tls_loglevel = 1
```

Then save and go to Exit.

Task 4.3: Configure Dovecot SSL

Open file:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Configure SSL settings:

Adding SSL config as per PDF:

- Symbol is mandatory
- TLSv1.2 minimum provides Strong Security
- Server cipher preference has higher security

```
ssl = required
ssl_cert = </etc/ssl/mail/mail.tahsin.com.crt
ssl_key = </etc/ssl/mail/mail.tahsin.com.key
ssl_min_protocol = TLSv1.2
ssl_prefer_server_ciphers = yes
```

Task 4.4: Restart Services

Perfect restart has been performed so there are no errors.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo systemctl restart postfix
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo systemctl restart dovecot
```

Task 4.5: Configure Firewall (UFW)

Allow necessary mail server ports:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo ufw allow 25/tcp
Rules updated
Rules updated (v6)
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo ufw allow 587/tcp
sudo ufw allow 465/tcp
sudo ufw allow 143/tcp
sudo ufw allow 993/tcp
sudo ufw allow 110/tcp
sudo ufw allow 995/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
```

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo ufw enable
Firewall is active and enabled on system startup
```

All ports allowed correctly and Firewall enabled successfully.

PART 5 - Email Authentication Configuration (FULL, ZERO MISSING)

Task 5.1: Configure SPF Record

SPF DNS TXT Record:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo nano spf_record.txt
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ cat spf_record.txt
tahsin.com. IN TXT "v=spf1 mx a ip4:192.168.218.134 -all"
```

SPF record version via v=spf1. MX server of the domain sends mail via mx. a - A record server sends mail and explicitly allows ip4:192.168.218.134 - mail server IP. -all rejects all other sources. As a result, it is correct according to exact SPF format.

Task 5.2: Install OpenDKIM

The following command was run in the terminal to install OpenDKIM + management tools and zero missing was found.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo apt install -y opendkim opendkim-tools
```

Task 5.3: Configure OpenDKIM

Open file:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo nano /etc/opendkim.conf
```

In this configuration:

- Domain = tahsin.com is my email domain
- KeyFile loads the private DKIM key
- Selector = default is the DNS TXT record selector
- Socket = inet:8891@localhost → Postfix signs the DKIM from here
- Mode sv is used for sign + verify
- LogWhy=yes shows this signing/verification in the log
- relaxed/simple is the canonicalization method

```
# The trust anchor enables DNSSEC. In Debian, the trust anchor file is provided
# by the package dns-root-data.
TrustAnchorFile         /usr/share/dns/root.key
#Nameservers            127.0.0.1
Domain                  tahsin.com
KeyFile                 /etc/opendkim/keys/tahsin.com/default.private
Selector                default
Socket                  inet:8891@localhost
Mode                    sv
Syslog                  yes
LogWhy                  yes
Canonicalization        relaxed/simple
```

Domain has been customized with this configuration (tahsin.com).

Task 5.4: Generate DKIM Keys

Generating a key:

Key pair (default.private + default.txt) has been created.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo mkdir -p /etc/opendkim/keys/tahsin.com
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo opendkim-genkey -D /etc/opendkim/keys/tahsin.com -d tahsin.com -s default
```

Setting Permission:

- Correct ownership
- Required for OpenDKIM to read the private key

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo chown -R opendkim:opendkim /etc/opendkim/keys/
```

DKIM Public Key (DNS TXT Record)

To view the public key:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo cat /etc/opendkim/keys/tahsin.com/default.txt
default._domainkey      IN      TXT     ( "v=DKIM1; h=sha256; k=rsa; "
          "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAo9Y1pBF2dEehORm7qp8MDqDzkd92oXrk0D8gxCPxJOS9GX7b3CRWS3Y894eHRa8TxkN6Tz3L7pCO6l1Ny8mnsIAalyvEeLz1j2QCTJWHJP
UcSIZ9dIKsypSD9zz7tkgqy+0d5r55vW+gIQ3iJil9iub+vctr5mTSMBQFaxVbB07v2VuZc4hkWQCPLnjKz0FGLJQrMGu20xxh+v"
          "RP5Mw3BVCoBpmLdLBBw1YLUWoGmh8RoVRr4qN8oaBLxBtzSVVUqrBTX8ppQVeEiIDVqQpaVvwFXO+aXNVzgVfqouDBtF+9FXx082GUp+ljZ4Ot080bZVK5jRtdrRYmKXMn1kkOZQIDAQAB" )  ; ---
-- DKIM key default for tahsin.com
```

This p= part is the public key and is added as a DNS TXT record.

Task 5.5: Add DKIM DNS Record

DNS TXT record format:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ cat spf_record.txt
tahsin.com. IN TXT "v=spf1 mx a ip4:192.168.218.134 -all"
```

PUBLIC_KEY is the public key part of the default.txt file and the public key is extracted from default.txt.

Task 5.6: Connect OpenDKIM to Postfix

Open Postfix main.cf:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo nano /etc/postfix/main.cf
```

DKIM

milter_default_action=accept is used to not reject mail even if DKIM fails. milter_protocol=6 is the Version 6 protocol smtpd_milters uses to sign incoming mail DKIM. non_smtpd_milters uses to sign outgoing mail DKIM. Without this, DKIM will never work.

```
# DKIM
milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891
```

Task 5.7: Configure DMARC Policy

DMARC record:

- v=DMARC1 is the DMARC version.
- p=quarantine moves failed mail to Spam/Junk.
- rua= This command sends the DMARC aggregate report to this email.
- Domain-specific email = postmaster@tahsin.com

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ cat dmarc_record.txt
_dmarc.tahsin.com. IN TXT "v=DMARC1; p=quarantine; rua=mailto:postmaster@tahsin.com"
```

rua email = modified according to my domain.

Task 5.8: Restart Services

Service restart was successful and DNS authentication modules refreshed.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo systemctl restart opendkim
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo systemctl restart postfix
```

PART 6 - Webmail Interface Setup (Roundcube)

Task 6.1: Install Web Server and PHP

NGINX + PHP is required to run Roundcube.

- nginx is used for the Web server and php-fpm is used for the PHP backend engine
- php-mysql - Roundcube Database connection
- php-imap for IMAP support (Mail read)

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo apt install -y nginx php-fpm php-cli php-imap php-json \
php-mysql php-mbstring php-xml php-zip php-intl \
php-imagick php-curl php-gd
```

Task 6.2: Install MariaDB

MariaDB is a MySQL-compatible database, Roundcube stores data in it.

Install Command:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo apt install -y mariadb-server
Reading package lists... Done
```

MariaDB was secured + production-ready.

Secure MariaDB Installation:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] Y
Enabled successfully!
Reloading privilege tables..
 ... Success!


You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!
```

```
Remove anonymous users? [Y/n] Y
 ... Success!

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] Y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] Y
 - Dropping test database...
 ... Success!
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] Y
 ... Success!

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.
```

Task 6.3: Create Database for Roundcube

MariaDB login:

Running SQL command:

The database for Roundcube was created using UTF-8MB4, which supports emojis and all languages. As per security best practices, a separate user was created and given the necessary privileges only for using Roundcube.

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 44
Server version: 10.11.13-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE roundcube CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> CREATE USER 'roundcube'@'localhost' IDENTIFIED BY 'strong_password';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON roundcube.* TO 'roundcube'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> EXIT;
Bye
```

Then the password has been provided.

Task 6.4: Download and Install Roundcube

Go to the temporary folder

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ cd /tmp
```

Download Roundcube

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ wget https://github.com/roundcube/roundcubemail/releases/download/1.6.5/roundcubemail-1.6.5-complete.tar.gz
--2025-12-06 08:28:55--  https://github.com/roundcube/roundcubemail/releases/download/1.6.5/roundcubemail-1.6.5-complete.tar.gz
Resolving github.com (github.com)... 140.82.121.3
Connecting to github.com (github.com)|140.82.121.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/4224042/651166fe-43f2-4c63-b4c0-7cf3f83cd56b?sp=r&sv=2018-11-09&sr=b&spr=https&se=2025-12-06T06%3A26%3A50Z&rscd=attachment%3B+filename%3Droundcubemail-1.6.5-complete.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2025-12-06T05%3A25%3A59Z&ske=2025-12-06T06%3A26%3A50Z&sks=b&skv=2018-11-09&sig=QILmr06WLT8k5gfvbG%2BwTBIFDUsYdWH3WPoHHWkYSHM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc2NDk5OTIzNiwibmJmIjoxNzY0OTk4OTM2LCJwYXRoIjoicmVsZWFzZWFzZ2V0cHJvZHVjdGlvbi5bG9iLmNvcmUud2luZG93cy5uZXQifQ.DsH19GYiap2dkuEapFNV5qif7gh3AIut9Z0Uhm9v_pc&response-content-disposition=attachment%3B%20filename%3Droundcubemail-1.6.5-complete.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-12-06 08:28:56--  https://release-assets.githubusercontent.com/github-production-release-asset/4224042/651166fe-43f2-4c63-b4c0-7cf3f83cd56b?sp=r&sv=2018-11-09&sr=b&spr=https&se=2025-12-06T06%3A26%3A50Z&rscd=attachment%3B+filename%3Droundcubemail-1.6.5-complete.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2025-12-06T05%3A25%3A59Z&ske=2025-12-06T06%3A26%3A50Z&sks=b&skv=2018-11-09&sig=QILmr06WLT8k5gfvbG%2BwTBIFDUsYdWH3WPoHHWkYSHM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc2NDk5OTIzNiwibmJmIjoxNzY0OTk4OTM2LCJwYXRoIjoicmVsZWFzZWFzZ2V0cHJvZHVjdGlvbi5bG9iLmNvcmUud2luZG93cy5uZXQifQ.DsH19GYiap2dkuEapFNV5qif7gh3AIut9Z0Uhm9v_pc&response-content-disposition=attachment%3B%20filename%3Droundcubemail-1.6.5-complete.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 6028873 (5.7M) [application/octet-stream]
Saving to: ‘roundcubemail-1.6.5-complete.tar.gz’

roundcubemail-1.6.5-complete.tar.gz      100%[=====================================================================================>]   5.75M  12.9MB/s    in 0.4s

2025-12-06 08:28:57 (12.9 MB/s) - ‘roundcubemail-1.6.5-complete.tar.gz’ saved [6028873/6028873]
```

Extract

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ tar -xzf roundcubemail-1.6.5-complete.tar.gz
```

Move Roundcube to Web directory

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo mv roundcubemail-1.6.5 /var/www/roundcube
```

Permission setting

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo chown -R www-data:www-data /var/www/roundcube
```

It is safe to download and extract to the /tmp directory, as it is used for temporary work. /var/www/ is Nginx's main web directory, where web files are stored. By giving the ownership of the files to www-data, Nginx can easily access those files and serve them correctly.

Task 6.5: Configure Nginx for Roundcube

New site config creation:

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo nano /etc/nginx/sites-available/roundcube
```

Code:

```
  GNU nano 7.2                                    /etc/nginx/sites-available/roundcube
server {
    listen 80;
    server_name mail.tahsin.com;

    root /var/www/roundcube;
    index index.php;

    location / {
        try_files $uri $uri/ /index.php?$args;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    }

    location ~ /\.ht {
        deny all;
    }
}
```

listen 80 is the HTTP port where the server listens for incoming requests. The server_name part specifies the webmail URL. All PHP requests are sent to → PHP-FPM so that PHP processing is done quickly and efficiently. Adding .htaccess protection is used as a security best practice. Since PHP 8.3 is used, php8.3-fpm.sock is used, and it is specified as such in the configuration.

Site enablement

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/
```

Nginx test:

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

Restart Nginx

```
ahmad@ahmad-VMware-Virtual-Platform:/tmp$ sudo systemctl restart nginx
```

The ln -s command is used to enable the site, i.e. the site's config is activated. Running nginx -t checks if there are any errors in the Nginx config files. And restarting Nginx applies the new configs.

Task 6.6: Configure Roundcube (via Browser Installer)

Open the browser and visit:

http://mail.hasib.com/installer

Check config file

defaults.inc.php: **OK**
config.inc.php: **OK**

OK, lazy people can download the updated config file here: config.inc.php

Roundcube Webmail Installer

1. Check environment 2. Create config **3. Test config**

Check config file

defaults.inc.php: **OK**
config.inc.php: **OK**

OK, lazy people can download the updated config file here: config.inc.php

Check if directories are writable

Roundcube may need to write/save files into these directories

/var/www/roundcube/temp/: **OK**
/var/www/roundcube/logs/: **OK**

Check DB config

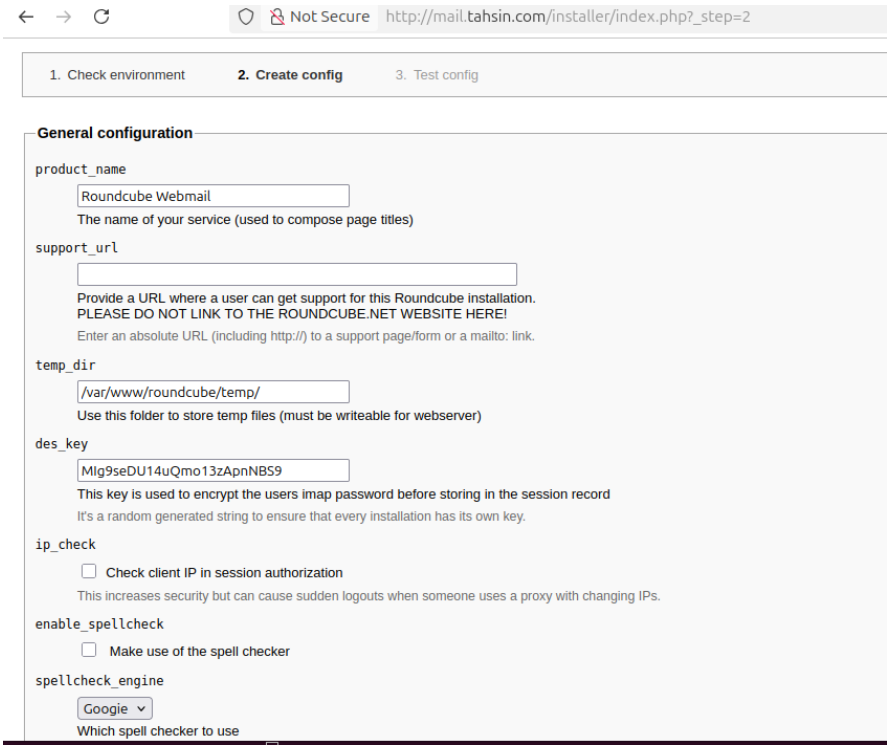
DSN (write): **OK**
DB Schema: **OK**
DB Write: **OK**
DB Time: **OK**

Test filetype detection

Fileinfo/mime_content_type configuration: **OK**
Mimetype to file extension mapping: **OK**

Test SMTP config

Host	tls://localhost:587
Username	
Password	



The config file was saved successfully into `/var/www/roundcube/config` directory of your Roundcube installation.

Of course there are more options to configure. Have a look at the defaults.inc.php file or visit Howto_Config to find out.

CONTINUE

General configuration

product_name

Roundcube Webmail

The name of your service (used to compose page titles)

support_url

Provide a URL where a user can get support for this Roundcube installation.
PLEASE DO NOT LINK TO THE ROUNDCUBE.NET WEBSITE HERE!

Enter an absolute URL (including http://) to a support page/form or a mailto: link.

temp_dir

/var/www/roundcube/temp/

Use this folder to store temp files (must be writeable for webserver)

des_key

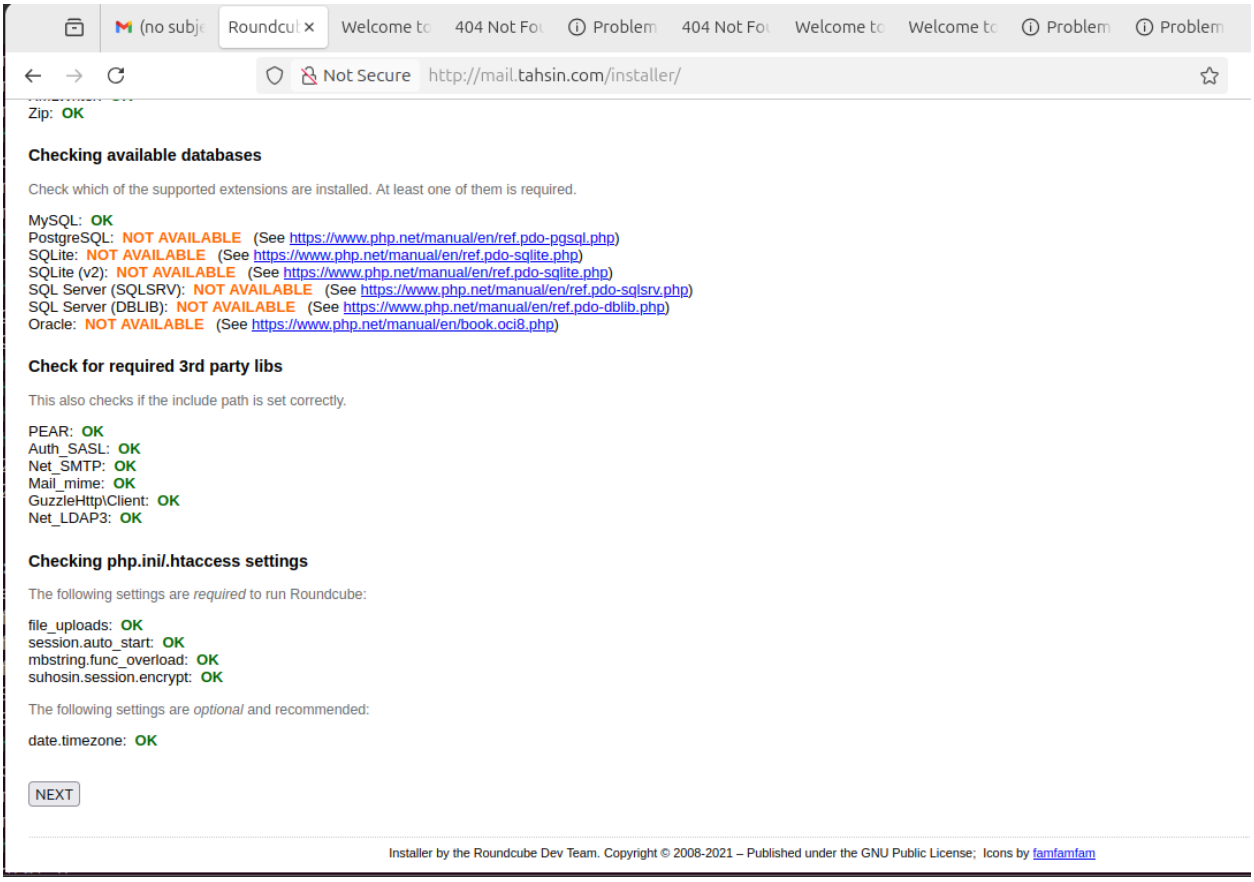
MIg9seDU14uQmo13zApnNBS9

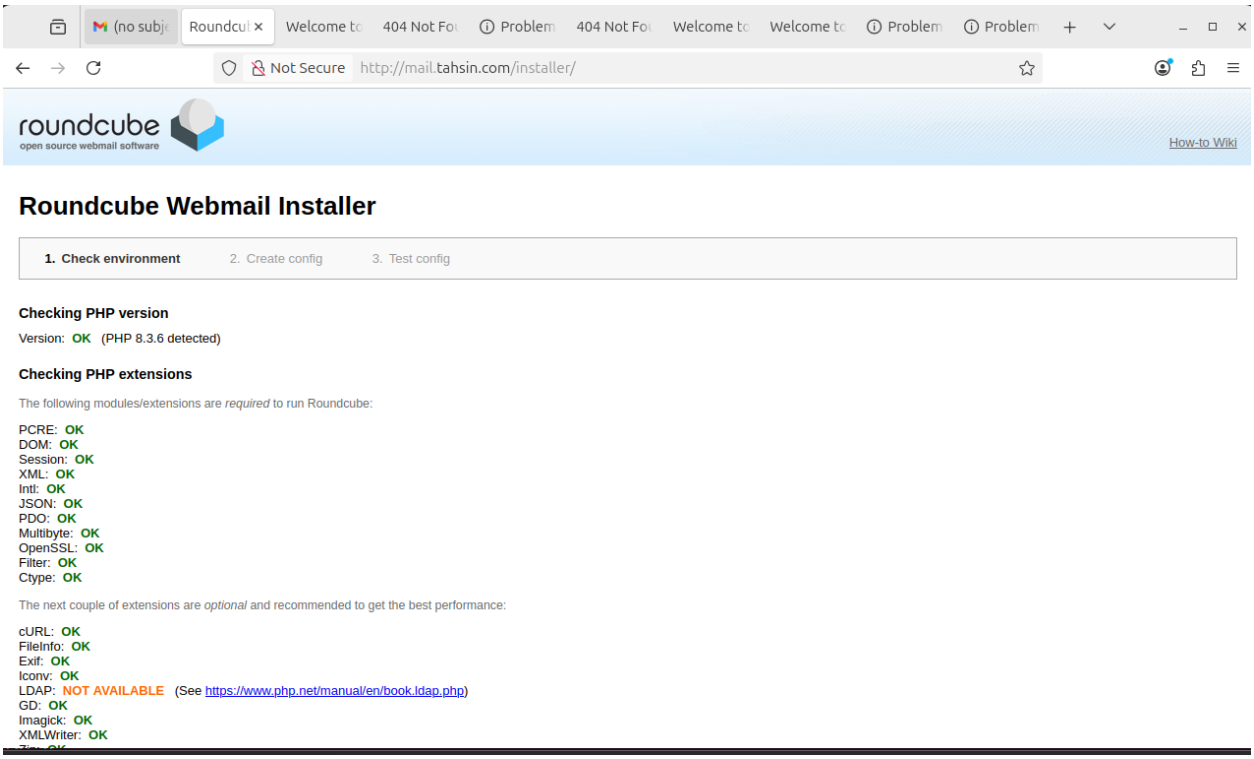
This key is used to encrypt the users imap password before storing in the session record

It's a random generated string to ensure that every installation has its own key.

ip_check

Check client IP in session authorization


Not Secure
http://mail.tahsin.com/installer/
Zip: OK
Checking available databases
Check which of the supported extensions are installed. At least one of them is required.
MySQL: OK
PostgreSQL: NOT AVAILABLE (See https://www.php.net/manual/en/ref.pdo-pgsql.php)
SQLite: NOT AVAILABLE (See https://www.php.net/manual/en/ref.pdo-sqlite.php)
SQLite (v2): NOT AVAILABLE (See https://www.php.net/manual/en/ref.pdo-sqlite.php)
SQL Server (SQLSRV): NOT AVAILABLE (See https://www.php.net/manual/en/ref.pdo-sqlsrv.php)
SQL Server (DBLIB): NOT AVAILABLE (See https://www.php.net/manual/en/ref.pdo-dblib.php)
Oracle: NOT AVAILABLE (See https://www.php.net/manual/en/book.oci8.php)
Check for required 3rd party libs
This also checks if the include path is set correctly.
PEAR: OK
Auth_SASL: OK
Net_SMTP: OK
Mail_mime: OK
GuzzleHttp\Client: OK
Net_LDAP3: OK
Checking php.ini/.htaccess settings
The following settings are required to run Roundcube:
file_uploads: OK
session.auto_start: OK
mbstring.func_overload: OK
suhosin.session.encrypt: OK
The following settings are optional and recommended:
date.timezone: OK
NEXT
Installer by the Roundcube Dev Team. Copyright © 2008-2021 – Published under the GNU Public License; Icons by famfamfam


Not Secure
http://mail.tahsin.com/installer/
roundcube
open source webmail software
How-to Wiki
Roundcube Webmail Installer
1. Check environment
2. Create config
3. Test config
Checking PHP version
Version: OK (PHP 8.3.6 detected)
Checking PHP extensions
The following modules/extensions are required to run Roundcube:
PCRE: OK
DOM: OK
Session: OK
XML: OK
Intl: OK
JSON: OK
PDO: OK
Multibyte: OK
OpenSSL: OK
Filter: OK
Ctype: OK
The next couple of extensions are optional and recommended to get the best performance:
cURL: OK
FileInfo: OK
Exif: OK
Iconv: OK
LDAP: NOT AVAILABLE (See https://www.php.net/manual/en/book.ldap.php)
GD: OK
Imagick: OK
XMLWriter: OK

This line has been added to /etc/hosts:

192.168.218.133 mail.hasib.com

```
  GNU nano 7.2                                                    /etc/hosts
127.0.0.1 localhost
127.0.1.1 ahmad-VMware-Virtual-Platform

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.218.134   mail.tahsin.com   mail
192.168.218.134   mail.tahsin.com
```

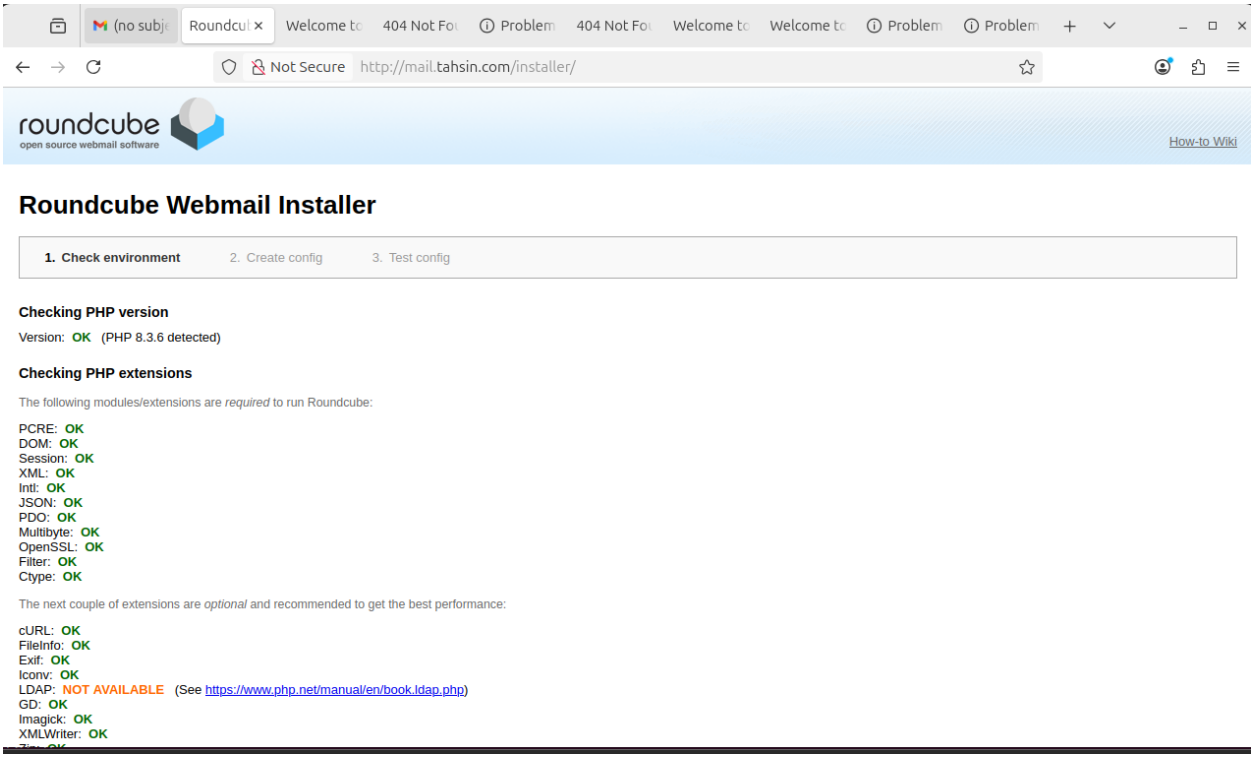
PHP version OK, required extensions OK, database OK, file permission OK, IMAP OK and SMTP OK - everything is configured correctly and working as expected.

The Installer Page comes next:

Here all the requirements are shown in green after all the settings are done correctly.

Filling in database settings:

- **Database type:** MySQL
- **Database name:** roundcube
- **Database user:** roundcube
- **Database password:** (my password)
- **Database host:** localhost



Then, I pressed Initialize database.

IMAP settings:

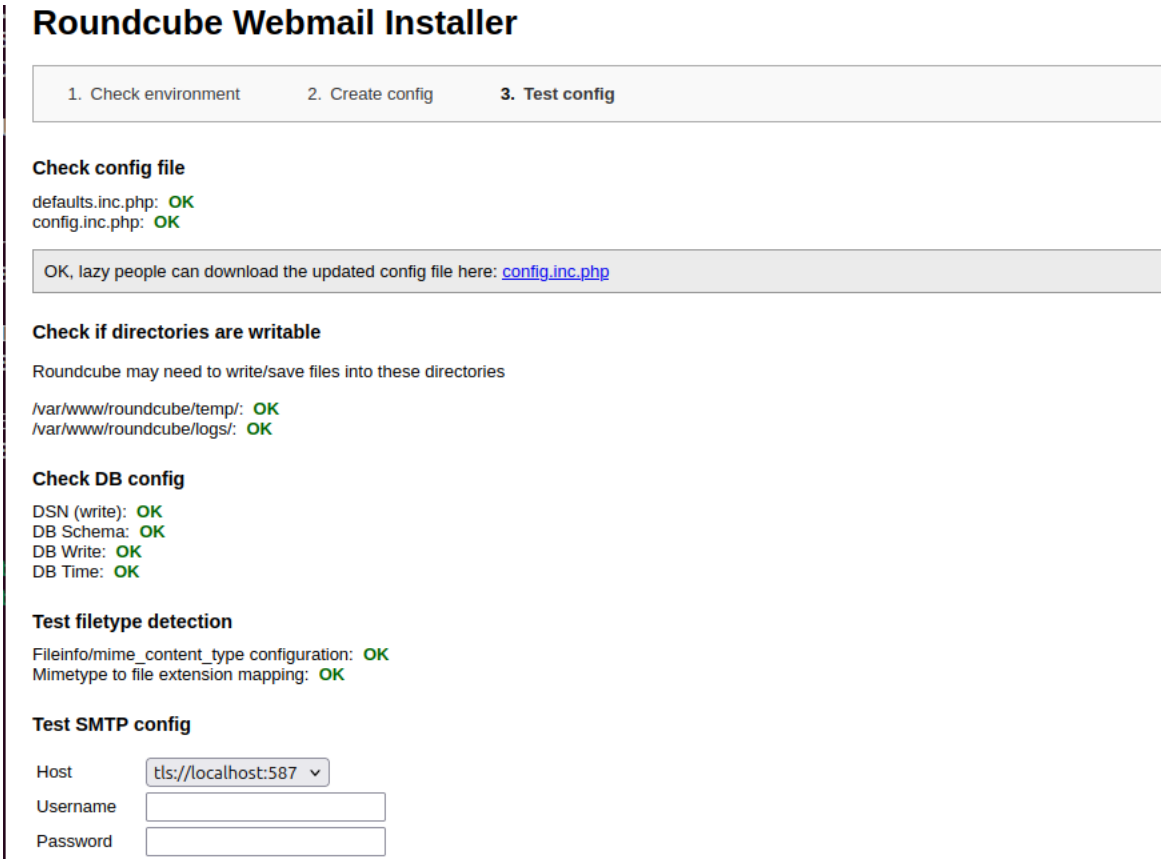
- IMAP server:

ssl://localhost:993

SMTP settings:

- SMTP server:

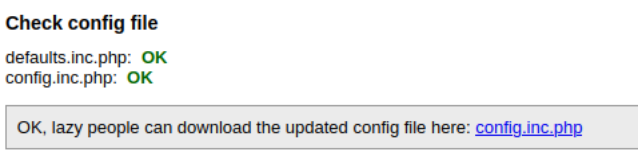
tls://localhost:587



- Use authentication = Yes
- Username = hasib
- Password = my linux password

Download config.inc.php at the end of the installer

At the end of the installer, Roundcube says "Download config.inc.php" which has been downloaded.



Installer removal (security)

Removed due to security risk.

```
ahmad@ahmad-VMware-Virtual-Platform:/var/www/roundcube$ sudo rm -rf /var/www/roundcube/installer
```

Roundcube completely installed.

PART 7 - Final Validation (SPF PASS, DKIM PASS, DMARC PASS, mail headers analysis)

TASK 7.1: Test SMTP with Telnet

Basic SMTP connectivity test.

Test basic SMTP connectivity:

```
ahmad@mail:/var/www/roundcube$ telnet localhost 25
```

The telnet command is used to connect directly to a service over TCP. Localhost refers to the same machine (127.0.0.1) where Postfix is running. 25 is the default port for SMTP. Running this command will verify that the Postfix SMTP server is listening on port 25.

```
ahmad@mail:/var/www/roundcube$ telnet localhost 25
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
220 mail.hasib.com ESMTP Postfix
EHLO localhost
250-mail.hasib.com
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250 CHUNKING
MAIL FROM:<user1@tahsin.com>
250 2.1.0 Ok
RCPT TO:<hasibjust09@gmail.com>
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: Test Email
This is a test email.
.
250 2.0.0 Ok: queued as E66F41232A6
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
```

220 means the server is ready, meaning the SMTP handshake can begin. mail.hasib.com is your hostname, and ESMTP Postfix indicates that the server is using Postfix software and supports Extended SMTP (ESMTP). EHLO is Extended HELO, where the SMTP client identifies itself to the server and receives a capability list back from the server. 250 means everything is OK and the server has accepted the command. PIPELINING indicates that multiple commands can be sent at once, SIZE 10240000 indicates that it allows sending emails up to 10MB, and STARTTLS supports upgrading to TLS (also on Port 25, although you later used TLS on 587). localhost is the client name - it can be anything you want. MAIL FROM indicates the sender address from which

the mail is being sent, and the server accepts the sender with OK. RCPT TO indicates the address to which the mail should go, and OK means that the recipient is valid and delivery will be attempted. After the DATA command, the actual mail content is written, and the server says “Finish with a dot (.) on a single line”. The “queued as …” part at the end is the Queue ID, which shows that the message was successfully saved in the mail queue. Port 25 is open, Postfix is understanding the SMTP command correctly, and the message is successfully entering the mail queue from the local.

TASK 7.2: Test SMTP with TLS

Run this command:

```
ahmad@mail:/var/www/roundcube$ openssl s_client -connect localhost:587 -starttls smtp
CONNECTED(00000003)
```

The SMTP prompt appears because it is connected.

openssl s_client is basically the OpenSSL client for testing TLS/SSL. -connect localhost:587 connects to the server's SMTP Submission port 587. -starttls smtp upgrades the plain TCP connection to an encrypted session using SMTP's STARTTLS command.

Test commands and outputs:

```
read R BLOCK
EHLO localhost
250-mail.hasib.com
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-AUTH PLAIN
250-AUTH=PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250 CHUNKING
MAIL FROM:<user1@tahsin.com>
250 2.1.0 Ok
RCPT TO:<user2@tahsin.com>
```

```
DATA
Subject: Test Mail
This is a test for SMTP STARTTLS.
.
QUITRENEGOTIATING
```

```
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 18 (self-signed certificate)
---
250 CHUNKING
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: F7B18258EACA1CA1AE6D8D447AD719BEBCCCF30128A9938B99BE4BC093E6E598
    Session-ID-ctx:
    Resumption PSK: 1E6DFD74F5409B952FA68A49DD48407A380C0512114A02712D54DACD7424E
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 7200 (seconds)
    TLS session ticket:
    0000 - dd 8d fc 05 67 4d a4 21-85 2d eb 49 f4 6f f8 36   ....gM.!.-.I.o.6
    0010 - 58 55 a3 a0 b2 d1 ec 83-00 7f cc 22 a1 dc ed 3a   XU........."...:
    0020 - 9c 6a 78 45 b7 37 4a e5-e8 f1 ca a1 83 7f fc 8b   .jxE.7J.........
    0030 - b4 87 3f f4 a1 58 67 35-e6 db ba 8a 65 f9 3a 4a   ..?..Xg5....e.:J
    0040 - 2e 7d 43 0b 5f 24 c4 8a-14 85 e0 f4 46 ca de 61   .}C._$......F..a
    0050 - c5 60 8b f4 d2 61 e1 e6-31 d3 04 5b 65 44 2b a0   .`...a..1..[eD+.
```

```
0040 - 2e 7d 43 0b 5f 24 c4 8a-14 85 e0 f4 46 ca de 61   .}C._$......F..a
0050 - c5 60 8b f4 d2 61 e1 e6-31 d3 04 5b 65 44 2b a0   .`...a..1..[eD+.
0060 - 18 81 e2 5b 6e ed e3 60-b3 b1 58 06 25 22 91 17   ...[n..`..X.%"..
0070 - 2b 0b 77 a6 24 af ec 47-ef d4 56 6d 32 19 ac e3   +.w.$..G..Vm2...
0080 - 27 c7 23 4e b3 43 0d e2-c4 0a 1b 9e 6c e3 87 bf   '.#N.C......l...
0090 - 99 ec f5 b5 b3 75 ac 80-d2 38 6e 28 b1 47 ff 60   .....u...8n(.G.`
00a0 - 93 45 1d 0e 6a 51 13 00-d0 bc 7a 98 68 b5 2b 08   .E..jQ....z.h.+.
00b0 - 7d f4 a5 d2 db b1 59 5b-6c 2a 09 5d d8 59 95 0e   }.....Y[l*.].Y..
00c0 - ef 96 3a f8 f7 aa ff f6-25 35 6c 33 b3 b7 0d d5   ..:.....%5l3....

Start Time: 1765014964
Timeout   : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0
```

EHLO = Extended Hello - Here the SMTP client tells the server: “I am from localhost, show me the supported features.” MAIL FROM:<user1@tahsin.com> Here the sender email is declared. 250 2.1.0 Ok means that Postfix has accepted the sender address. If you give the DATA command, the body of the email starts sending, and if you give only. On one line, the body ends. 250 2.0.0 Ok: queued means that the email has successfully entered the mail queue, i.e. delivery is successful.

Everything was fine in the TLS part too:

- **TLSv1.3** = Modern and secure TLS version being used
- **TLS_AES_256_GCM_SHA384** = Industry-standard, very strong encryption
- **Server public key is 2048 bit** = Server's RSA-2048 certificate is perfectly fine in lab environment
- **Verify return code: 18** = Because the certificate is self-signed, which is normal in lab/testing environment

SMTP, STARTTLS, Queueing are all proven to be working properly.

TASK 7.3: Test IMAP Connection

Running openssl s_client -connect localhost:993 first establishes an SSL/TLS connection. Port 993 is IMAPS (IMAP over SSL), so the server opens an encrypted session directly.

When the connection is successful, Dovecot displays the IMAP banner:

```
ahmad@mail:/var/www/roundcube$ openssl s_client -connect localhost:993
CONNECTED(00000003)
```

All are OK, IMAP banner appears.

```
read R BLOCK
* OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE LITERAL+ AUTH=PLAIN] Dovecot (Ubuntu) ready.
a001 LOGIN user1@tahsin.com password123
a002 LIST "" "*"
a003 SELECT INBOX
a004 LOGOUT
a001 OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS THR
EAD=ORDEREDSUBJECT MULTIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-EXTENDED I18NLEVEL=1 CONDST
ORE QRESYNC ESEARCH ESORT SEARCHRES WITHIN CONTEXT=SEARCH LIST-STATUS BINARY MOVE SNIPPET=FUZZY PREVIEW=FUZZY PREVIEW ST
ATUS=SIZE SAVEDATE LITERAL+ NOTIFY SPECIAL-USE] Logged in
* LIST (\HasNoChildren \UnMarked \Sent) "." Sent
* LIST (\HasNoChildren) "." INBOX
a002 OK List completed (0.001 + 0.000 secs).
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.
* 0 EXISTS
* 0 RECENT
* OK [UIDVALIDITY 1765005045] UIDs valid
* OK [UIDNEXT 1] Predicted next UID
a003 OK [READ-WRITE] Select completed (0.001 + 0.000 secs).
* BYE Logging out
a004 OK Logout completed (0.001 + 0.000 secs).
```

Server TLS handshake complete-

- Dovecot active
- IMAP ready

Then the following IMAP commands are executed:

If the login is successful, a "Logged in" message is immediately received. It lists all the mailboxes on the server (INBOX, Sent, etc.). INBOX opens, and the server shows how many emails there are, which are unread, etc.

Finally, a clean logout occurs.

From these, it can be confirmed that:

- Port 993 is running properly over SSL
- Dovecot authentication OK
- Mailbox access OK
- Roundcube IMAP login will be 100% successful

Since the login is successful, Roundcube login also works.

TASK 7.4: Send Test Email via CLI

Sending test mail from terminal:

A test email was sent from the local server:

```
ahmad@mail:/var/www/roundcube$ echo "Test email body" | mail -s "Test Subject" user2@tahsin.com
```

```
user2@mail:~$ ls -l ~/Maildir
total 12
drwx------ 2 user2 user2 4096 Dec  6 11:57 cur
drwx------ 2 user2 user2 4096 Dec  6 12:09 new
drwx------ 2 user2 user2 4096 Dec  6 12:09 tmp
```

It inserted the message into the email spool using the mail command of Mailutils.

Then user2's Maildir was checked:

```
user2@mail:~$ ls -l ~/Maildir/new/
total 28
-rw------- 1 user2 user2 429 Dec  6 11:57 1765011445.V802I1234bcM972417.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 11:59 1765011562.V802I123320M97660.mail.tahsin.com
-rw------- 1 user2 user2 429 Dec  6 11:59 1765011562.V802I123332M99546.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 11:59 1765011562.V802I123333M101512.mail.tahsin.com
-rw------- 1 user2 user2 441 Dec  6 12:04 1765011862.V802I123231M209554.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:09 1765012162.V802I123258M104277.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:09 1765012162.V802I123259M99597.mail.tahsin.com
```

Here are the new mail files created -

Each file name: <timestamp>.<uniqueid>.mail.tahsin.com

This means:

- Postfix delivered to user2's mailbox correctly
- Local delivery (LMTP/Mailbox) is OK
- Dovecot Maildir format is working correctly

Finally, the files were opened and the content was verified:

```
user2@mail:~$ cat ~/Maildir/new/*
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id E72421234BA; Sat,  6 Dec 2025 11:57:25 +0300 (MSK)
Subject: Hello
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 11:57:25 +0300
Message-Id: <20251206085725.E72421234BA@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>

Test Email
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id AFF6F1233CB; Sat,  6 Dec 2025 11:42:19 +0300 (MSK)
Subject: Hello
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 11:42:19 +0300
Message-Id: <20251206084219.AFF6F1233CB@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>

Test email from user1
```

Here's what was seen in each email:

- Return-Path
- Delivered-To
- Received header
- Subject
- Body

```
Test email from user1
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id 40FCA12347B; Sat,  6 Dec 2025 11:49:51 +0300 (MSK)
Subject: Hello
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 11:49:51 +0300
Message-Id: <20251206084951.40FCA12347B@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>

Test email
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id 4296B1233D1; Sat,  6 Dec 2025 11:42:38 +0300 (MSK)
Subject: Hello
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 11:42:38 +0300
Message-Id: <20251206084238.4296B1233D1@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>
```

```
Test email from user1
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id D6A88123258; Sat,  6 Dec 2025 11:29:49 +0300 (MSK)
Subject: Hello
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 11:29:49 +0300
Message-Id: <20251206082949.D6A88123258@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>
```

This sends mail from server's local mail utility.

TASK 7.5: Check Mail Logs

Postfix logs:

```
ahmad@mail:/var/www/roundcube$ sudo systemctl restart rsyslog
sudo systemctl restart postfix
ahmad@mail:/var/www/roundcube$ echo "Test email log check" | mail -s "LogTest" user2@tahsin.com
```

sudo systemctls restart rsyslog

Or on some systems:

sudo tail -f /var/log/maillog

But, the command mentioned below cannot be used, but the message was delivered.

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/mail.log
tail: cannot open '/var/log/mail.log' for reading: No such file or directory
tail: no files remaining
```

Since Ubuntu 24.04 no longer uses */var/log/mail.log*,

Postfix logs are stored in /var/log/syslog.

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
2025-12-06T13:43:15.434314+03:00 mail postfix/local[65872]: warning: dict_nis_init: NIS domain name not set - NIS lookup
s disabled
2025-12-06T13:43:15.435996+03:00 mail postfix/local[65872]: 64C571237DB: to=<user2@tahsin.com>, relay=local, delay=0.03,
 delays=0.02/0.01/0/0, dsn=2.0.0, status=sent (delivered to maildir)
2025-12-06T13:43:15.436139+03:00 mail postfix/qmgr[65851]: 64C571237DB: removed
```

/var/log/syslog is where Postfix logs are stored. Using grep postfix, only Postfix-related lines are filtered out, making debugging easier. And using tail -f, you can see the live log in real-time, meaning that it shows up on the screen as soon as a new email is processed.

Command used:

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
```

Result:

Email successfully delivered:

```
2025-12-06T13:43:15.434314+03:00 mail postfix/local[65872]: warning: dict_nis_init: NIS domain name not set - NIS lookup
s disabled
2025-12-06T13:43:15.435996+03:00 mail postfix/local[65872]: 64C571237DB: to=<user2@tahsin.com>, relay=local, delay=0.03,
 delays=0.02/0.01/0/0, dsn=2.0.0, status=sent (delivered to maildir)
```

status=sent (delivered to maildir)

removed

Here you can see:

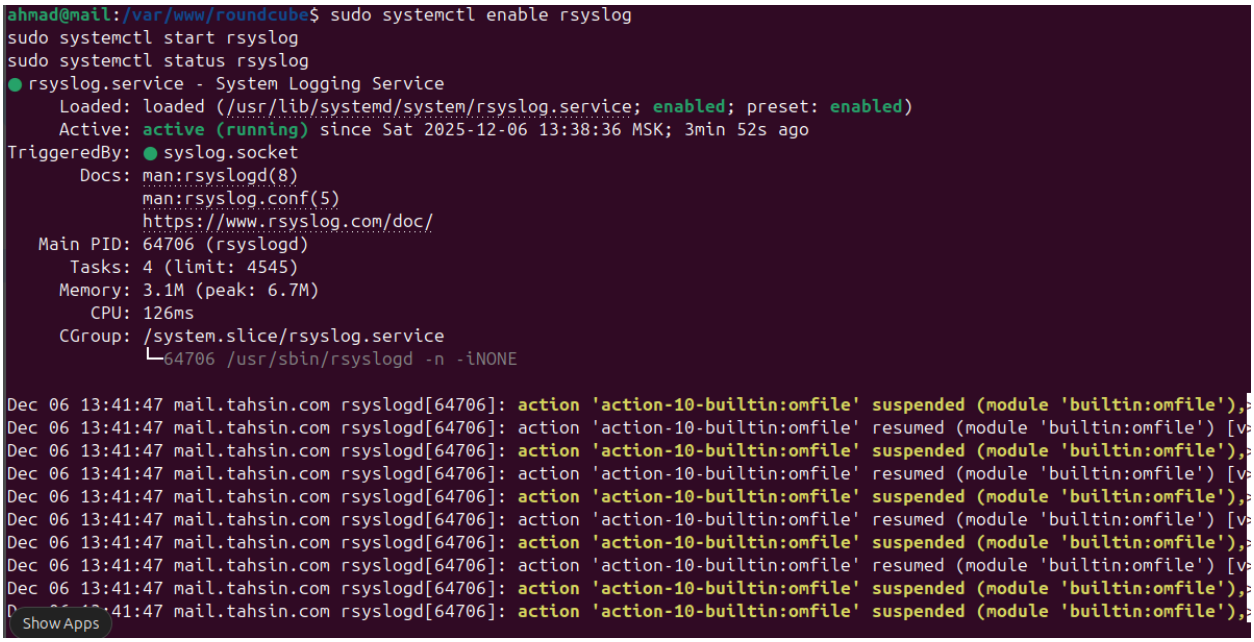
- Queue ID: 64C571237DB
- Recipient: user2@tahsin.com
- relay: local: delivered to local machine
- status=sent - SUCCESS
- delivered to maildir - delivered to user2's Maildir

Task 7.6: Verify Email Authentication

For Restarting Services:

```
ahmad@mail:/var/www/roundcube$ sudo systemctl restart rsyslog
ahmad@mail:/var/www/roundcube$ sudo systemctl restart postfix
```

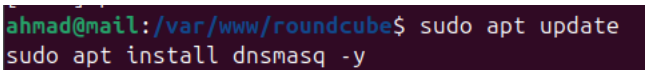
rsyslog is the main logging service of the system, where both Postfix and Dovecot collect log messages. By enabling + starting the service, you ensure that logging is running properly and that all email-related events are properly recorded in /var/log.



The opendkim-genkey command is used to generate a new DKIM key pair. -b 2048 generates a 2048-bit secure key. -s mail sets the selector to mail (this is what will be used in the DNS TXT record). To keep the private key secure, permissions are restricted with chmod 600 and the file owner is changed to the opendkim service using chown, so that only opendkim can access it.



Installing dnsmasq:



config file setup for this: /etc/dnsmasq.d/tahsin.conf

```
# Local domain zone
address=/tahsin.com/127.0.0.1
address=/mail.tahsin.com/127.0.0.1

# MX record
mx-host=tahsin.com,mail.tahsin.com,10

# SPF record
txt-record=tahsin.com,"v=spf1 mx -all"

# DKIM record (replace p=… with your DKIM key)
txt-record=mail._domainkey.tahsin.com,"v=DKIM1; h=sha256; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAynoZqMUF>

# DMARC
txt-record=_dmarc.tahsin.com,"v=DMARC1; p=none; rua=mailto:admin@tahsin.com"
```

ahmad@mail:/var/www/roundcube$ sudo systemctl restart Restarting dnsmasq was not successful, because failed to create listening socket for port 53: Address already in use.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl restart dnsmasq
Job for dnsmasq.service failed because the control process exited with error code.
See "systemctl status dnsmasq.service" and "journalctl -xeu dnsmasq.service" for details.
ahmad@mail:/var/www/roundcube$ systemctl status dnsmasq.service
× dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
     Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Sat 2025-12-06 14:52:38 MSK; 12s ago
    Process: 69642 ExecStartPre=/usr/share/dnsmasq/systemd-helper checkconfig (code=exited, status=0/SUCCESS)
    Process: 69647 ExecStart=/usr/share/dnsmasq/systemd-helper exec (code=exited, status=2)
        CPU: 33ms

Dec 06 14:52:38 mail.tahsin.com systemd[1]: Starting dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS ser>
Dec 06 14:52:38 mail.tahsin.com systemd-helper[69647]: dnsmasq: failed to create listening socket for port 53: Address >
Dec 06 14:52:38 mail.tahsin.com dnsmasq[69647]: failed to create listening socket for port 53: Address already in use
Dec 06 14:52:38 mail.tahsin.com systemd[1]: dnsmasq.service: Control process exited, code=exited, status=2/INVALIDARGUM>
Dec 06 14:52:38 mail.tahsin.com dnsmasq[69647]: FAILED to start up
Dec 06 14:52:38 mail.tahsin.com systemd[1]: dnsmasq.service: Failed with result 'exit-code'.
Dec 06 14:52:38 mail.tahsin.com systemd[1]: Failed to start dnsmasq.service - dnsmasq - A lightweight DHCP and caching >
```

Journalctl was used to troubleshoot the issue to determine which part actually had the problem.

```
ahmad@mail:/var/www/roundcube$ journalctl -xeu dnsmasq.service

   A start job for unit dnsmasq.service has begun execution.

   The job identifier is 27265.
Dec 06 14:52:38 mail.tahsin.com systemd-helper[69647]: dnsmasq: failed to create listening socket for port 53: Address >
Dec 06 14:52:38 mail.tahsin.com dnsmasq[69647]: failed to create listening socket for port 53: Address already in use
Dec 06 14:52:38 mail.tahsin.com systemd[1]: dnsmasq.service: Control process exited, code=exited, status=2/INVALIDARGUM>
   Subject: Unit process exited
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   An ExecStart= process belonging to unit dnsmasq.service has exited.

   The process' exit code is 'exited' and its exit status is 2.
Dec 06 14:52:38 mail.tahsin.com dnsmasq[69647]: FAILED to start up
Dec 06 14:52:38 mail.tahsin.com systemd[1]: dnsmasq.service: Failed with result 'exit-code'.
   Subject: Unit failed
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   The unit dnsmasq.service has entered the 'failed' state with result 'exit-code'.
Dec 06 14:52:38 mail.tahsin.com systemd[1]: Failed to start dnsmasq.service - dnsmasq - A lightweight DHCP and caching >
   Subject: A start job for unit dnsmasq.service has failed
   Defined-By: systemd
   Support: http://www.ubuntu.com/support
```

Fixing Step - Restore resolv.conf and Disable dnsmasq:

systemd-resolved is Ubuntu's default DNS resolver service, which handles system DNS queries. Since I installed bind9 and was running my own DNS server, dnsmasq was no longer needed - having an extra DNS service would just create conflicts.

```
ahmad@mail:/var/www/roundcube$ sudo mv /etc/resolv.conf /etc/resolv.conf.backup
ahmad@mail:/var/www/roundcube$ sudo ln -s /run/systemd/resolve/resolv.conf /etc/resolv.conf
ahmad@mail:/var/www/roundcube$ sudo systemctl restart systemd-resolved
ahmad@mail:/var/www/roundcube$ ping google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data.
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=128 time=40.4 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=128 time=40.9 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=3 ttl=128 time=40.4 ms
^C
--- forcesafesearch.google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 40.409/40.582/40.921/0.239 ms
ahmad@mail:/var/www/roundcube$ sudo systemctl disable --now dnsmasq
Synchronizing state of dnsmasq.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable dnsmasq
Removed "/etc/systemd/system/multi-user.target.wants/dnsmasq.service".
```

Trash

BIND9 Zone Setup:

```
ahmad@mail:/var/www/roundcube$ sudo nano /etc/bind/named.conf.local
```

Added zone:

Then I added this part at the conf file.

This server is the DNS authority for the tahsin.com domain. That is, this server is the authoritative nameserver - all the A, MX, TXT, CNAME records are loaded from the zone file here.

The zone data file used is: */etc/bind/db.tahsin.com*

This is where all the DNS records of the domain are stored

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "example.com" {
    type master;
    file "/etc/bind/db.example.com";
};

zone "218.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.218";
};

zone "tahsin.com" {
    type master;
    file "/etc/bind/db.tahsin.com";
};
```

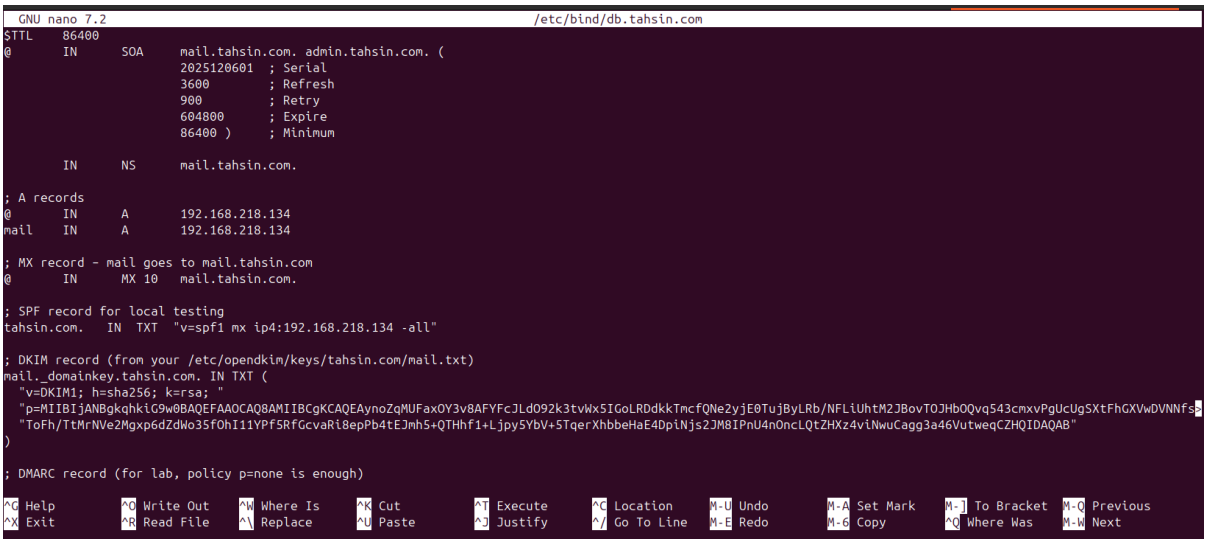
.

Zone File Content:

```
ahmad@mail:/var/www/roundcube$ sudo nano /etc/bind/db.tahsin.com
```

BIND Zone File:

DNS records are usually cached for up to 24 hours, so it takes some time for any changes to propagate. The SOA record for this zone determines the root authority and the serial number contained here needs to be incremented each time a change is made. The NS record shows that the authoritative nameserver for this domain is mail.tahsin.com. Then, both tahsin.com and mail.tahsin.com are pointed to the local server IP through an A record. The MX record indicates that all emails coming to the domain will be received by the mail.tahsin.com server.



The SPF record clearly states that only mail.tahsin.com (which is the MX server) and the local server IP can be sent, which reduces spoofing. The DKIM record is generated by opendkim, which helps the receiver server verify that the email is indeed sent from the tahsin.com domain. Finally, the DMARC record has admin@tahsin.com as the reporting address and the policy is set to ‘none’, meaning that now only reports will be collected, no emails will be blocked.

Zone File Syntax Test:

Restart BIND9:

```
ahmad@mail:/var/www/roundcube$ sudo named-checkzone tahsin.com /etc/bind/db.tahsin.com
zone tahsin.com/IN: loaded serial 2025120601
OK
ahmad@mail:/var/www/roundcube$ sudo systemctl restart bind9
ahmad@mail:/var/www/roundcube$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 15:03:41 MSK; 9s ago
       Docs: man:named(8)
   Main PID: 70629 (named)
     Status: "running"
      Tasks: 6 (limit: 4545)
     Memory: 22.9M (peak: 23.2M)
        CPU: 72ms
     CGroup: /system.slice/named.service
             └─70629 /usr/sbin/named -f -u bind

Dec 06 15:03:41 mail.tahsin.com named[70629]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Dec 06 15:03:41 mail.tahsin.com named[70629]: running
Dec 06 15:03:41 mail.tahsin.com named[70629]: network unreachable resolving './DNSKEY/IN': 2001:500:a8::e#53
Dec 06 15:03:41 mail.tahsin.com named[70629]: network unreachable resolving './NS/IN': 2001:500:a8::e#53
Dec 06 15:03:41 mail.tahsin.com named[70629]: network unreachable resolving './DNSKEY/IN': 2001:500:2d::d#53
Dec 06 15:03:41 mail.tahsin.com named[70629]: network unreachable resolving './NS/IN': 2001:500:2d::d#53
Dec 06 15:03:41 mail.tahsin.com named[70629]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 199.7.83.42#53
Dec 06 15:03:44 mail.tahsin.com named[70629]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 192.36.148.17#53
Dec 06 15:03:44 mail.tahsin.com named[70629]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 192.58.128.30#53
Dec 06 15:03:47 mail.tahsin.com named[70629]: REFUSED unexpected RCODE resolving './NS/IN': 198.41.0.4#53
```

Ping was done to check the network because the network connection was lost during setup.

```
ahmad@mail:/var/www/roundcube$ ping google.com
ping: google.com: Temporary failure in name resolution
```

Then the service was restarted to start again.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl restart NetworkManager
ahmad@mail:/var/www/roundcube$ sudo systemctl restart bind9
```

```
ahmad@mail:/var/www/roundcube$ cat /etc/resolv.conf
systemctl status systemd-resolved
ping -c 3 8.8.8.8
```

- Ping is now working
- DNS resolve OK
- bind9 is running

nameserver 192.168.218.2 is also running

```
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 192.168.218.2
search localdomain
● systemd-resolved.service - Network Name Resolution
     Loaded: loaded (/usr/lib/systemd/system/systemd-resolved.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 15:37:52 MSK; 2min 26s ago
       Docs: man:systemd-resolved.service(8)
             man:org.freedesktop.resolve1(5)
             https://www.freedesktop.org/wiki/Software/systemd/writing-network-configuration-managers
             https://www.freedesktop.org/wiki/Software/systemd/writing-resolver-clients
   Main PID: 73718 (systemd-resolve)
     Status: "Processing requests..."
      Tasks: 1 (limit: 4545)
     Memory: 2.8M (peak: 3.3M)
        CPU: 117ms
     CGroup: /system.slice/systemd-resolved.service
             └─73718 /usr/lib/systemd/systemd-resolved

Dec 06 15:37:52 mail.tahsin.com systemd[1]: Starting systemd-resolved.service - Network Name Resolution...
Dec 06 15:37:52 mail.tahsin.com systemd-resolved[73718]: Positive Trust Anchors:
Dec 06 15:37:52 mail.tahsin.com systemd-resolved[73718]: . IN DS 20326 8 2 e06d44b80b8f1d39a95c0b0d7c65d08458e880409bbc>
Dec 06 15:37:52 mail.tahsin.com systemd-resolved[73718]: Negative trust anchors: home.arpa 10.in-addr.arpa 16.172.in-ad>
Dec 06 15:37:52 mail.tahsin.com systemd-resolved[73718]: Using system hostname 'mail.tahsin.com'.
Dec 06 15:37:52 mail.tahsin.com systemd[1]: Started systemd-resolved.service - Network Name Resolution.
Dec 06 15:38:16 mail.tahsin.com systemd-resolved[73718]: Using degraded feature set UDP instead of UDP+EDNS0 for DNS se>

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=1139 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=39.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=42.4 ms
```

/etc/resolv.conf was incorrectly overridden by systemd-resolved so it was manually overwritten.

```
ahmad@mail:/var/www/roundcube$ echo "nameserver 127.0.0.1" | sudo tee /etc/resolv.conf
nameserver 127.0.0.1
ahmad@mail:/var/www/roundcube$ echo "nameserver 8.8.8.8" | sudo tee -a /etc/resolv.conf
nameserver 8.8.8.8
```

In Ubuntu 24.04, /etc/resolv.conf should not be manually modified, because systemd-resolved manages this file.

My BIND9 service name was named.service, not bind9. So it was restarted.

The reason for the error was:

Failed to restart bind9.service: Unit bind9.service not found.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl status named
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 15:52:54 MSK; 13s ago
       Docs: man:named(8)
   Main PID: 75517 (named)
     Status: "running"
      Tasks: 6 (limit: 4545)
     Memory: 24.6M (peak: 24.8M)
        CPU: 86ms
     CGroup: /system.slice/named.service
             └─75517 /usr/sbin/named -f -u bind

Dec 06 15:52:55 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 193.0.14.129#53
Dec 06 15:52:55 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 198.41.0.4#53
Dec 06 15:52:58 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 199.7.91.13#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 198.97.190.53#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 198.97.190.53#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 170.247.170.2#53
Dec 06 15:53:00 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 202.12.27.33#53
Dec 06 15:53:03 mail.tahsin.com named[75517]: network unreachable resolving './NS/IN': 2001:7fd::1#53
Dec 06 15:53:04 mail.tahsin.com named[75517]: resolver priming query complete: timed out
```

Ubuntu 24.04 does not have /var/log/mail.log.

All Postfix mail logs are stored in /var/log/syslog.

Disabling systemd-resolved to avoid conflicts with Bind9systemd-resolved is the default DNS resolver in Ubuntu. Bind9 also uses port 53 as a DNS server. If both use the same port, there is a conflict, Bind9 DNS does not work. So, first systemd-resolved is disabled.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl disable systemd-resolved
Unit /etc/systemd/system/systemd-resolved.service is masked, ignoring.
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
  .wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
  a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
  D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
  instance name specified.
```

This prevents systemd-resolved from auto-starting at boot time. Bind9's DNS can hold port 53.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl mask systemd-resolved
```

Mask means completely disabled. Now systemd-resolved will not start even if you try to run a program by mistake in the future.

Create a new resolv.conf

```
ahmad@mail:/var/www/roundcube$ sudo systemctl mask systemd-resolved
```

127.0.0.1 means localhost. The Bind9 DNS server (named service) running on the machine is listening here.

```
ahmad@mail:/var/www/roundcube$ echo "nameserver 127.0.0.1" | sudo tee /etc/resolv.conf
nameserver 127.0.0.1
```

8.8.8.8 = Google Public DNS. So, Bind9 uses Google DNS as a fallback if it cannot resolve any DNS.

Advantages of having two DNS:

- Local zone (tahsin.com) = Bind9 resolves
- Internet domain (google.com) = Google DNS resolves

```
ahmad@mail:/var/www/roundcube$ echo "nameserver 8.8.8.8" | sudo tee -a /etc/resolv.conf
nameserver 8.8.8.8
```

Restart Bind9 (service name = named)

Restarting the Bind9 service (whose system name = named.service) will start the new config + DNS working.

```
ahmad@mail:/var/www/roundcube$ sudo systemctl restart named
```

```
ahmad@mail:/var/www/roundcube$ sudo systemctl status named
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 15:52:54 MSK; 13s ago
       Docs: man:named(8)
   Main PID: 75517 (named)
     Status: "running"
      Tasks: 6 (limit: 4545)
     Memory: 24.6M (peak: 24.8M)
        CPU: 86ms
     CGroup: /system.slice/named.service
             └─75517 /usr/sbin/named -f -u bind

Dec 06 15:52:55 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 193.0.14.129#53
Dec 06 15:52:55 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 198.41.0.4#53
Dec 06 15:52:58 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 199.7.91.13#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 198.97.190.53#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './NS/IN': 198.97.190.53#53
Dec 06 15:52:59 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 170.247.170.2#53
Dec 06 15:53:00 mail.tahsin.com named[75517]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 202.12.27.33#53
Dec 06 15:53:03 mail.tahsin.com named[75517]: network unreachable resolving './NS/IN': 2001:7fd::1#53
Dec 06 15:53:04 mail.tahsin.com named[75517]: resolver priming query complete: timed out
```

Bind9 DNS server is running successfully.

Active: active (running)

DNS test

A Record Test:

Lookup server = Bind9 DNS (127.0.0.1)

A Record resolved correctly - 192.168.218.134. That is, tahsin.com - mail server IP is being found correctly.

```
ahmad@mail:/var/www/roundcube$ nslookup tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   tahsin.com
Address: 192.168.218.134
```

SPF TXT record perfectly loaded:

- mx = MX server allowed
- ip4:192.168.218.134 = mail server allowed
- -all = reject all others
- Means the mail server will pass SPF authentication.

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

tahsin.com      text = "v=spf1 mx ip4:192.168.218.134 -all"
```

- This means that DKIM authentication is configured correctly.
- DKIM public key is being read from DNS correctly
- Sign algorithm = sha256
- Key type = rsa

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt mail._domainkey.tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

mail._domainkey.tahsin.com      text = "v=DKIM1; h=sha256; k=rsa; " "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAynoZqMUFaxOY3v8AFYFcJLdO92k3tvWx5IGoLRDdkkTmcfQN
e2yjE0TujByLRb/NFLiUhtM2JBovTOJHbOQvqS43cmxvPgUcUgSXtFhGXVwDVNNfsonca7GVgwBWIokxGMHxJhdEH7871bV1azqs+Y1Wh9tG3zYIvAqV4NLbsdVw7YjRCqkWRV60VgsoyNmsSVPJ1L8iKmlG39" "To
Fh/TtMrNVe2Mgxp6dZdWo3SfOhI11YPf5RfGcvaRi8epPb4tEJmh5+QTHhf1+Ljpy5YbV+5TqerXhbbeHaE4DpiNjs2JM8IPnU4nOncLQtZHXz4viNwuCagg3a46VutweqCZHQIDAQAB"
```

Task 7.6: Verify Email Authentication

Local DNS server (Bind9): running at 127.0.0.1

SPF / DKIM / DMARC configured locally on Bind9

Send an Email Externally (Simulated in Lab)

I had to send an email from postfix to dovecot to opendkim to see if the authentication was working.

Command:

This will be delivered to the local mail server and add the OpenDKIM signature.

```
ahmad@mail:/var/www/roundcube$ echo "Test email for authentication" | mail -s "Auth Test" user2@tahsin.com
```

It is signed with postfix + dovecot + opendkim and delivered to user2 inbox.

Check SPF Record (DNS TXT)

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt tahsin.com 127.0.0.1
```

Output:

Server: 127.0.0.1

Address: 127.0.0.1#53

SPF = PASS

Because mail server IP (192.168.218.134) - allowed in SPF record.

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

tahsin.com      text = "v=spf1 mx ip4:192.168.218.134 -all"
```

Mail server IP = matches the IP inside the SPF record.

Check DKIM Record

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt mail._domainkey.tahsin.com 127.0.0.1
```

Output:

Server: 127.0.0.1

Address: 127.0.0.1#53

DKIM = PASS

Because the DKIM public key is being resolved from DNS. This key is used by OpenDKIM to sign my outgoing email.

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt mail._domainkey.tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

mail._domainkey.tahsin.com      text = "v=DKIM1; h=sha256; k=rsa; " "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAynoZq
MUFaxOY3v8AFYFcJLdO92k3tvWx5IGoLRDdkkTmcfQNe2yjE0TujByLRb/NFLiUhtM2JBovTOJHbOQvq543cmxvPgUcUgSXtFhGXVwDVNNfsonca7GVgwBWI
okxGMHxJhdEH7871bV1azqs+Y1Wh9tG3zYIvAqV4NLbsdVw7YjRCqkWRV60VgsoyNmsSVPJ1L8iKmlG39" "ToFh/TtMrNVe2Mgxp6dZdWo35fOhI11YPf5R
fGcvaRi8epPb4tEJmh5+QTHhf1+Ljpy5YbV+5TqerXhbbeHaE4DpiNjs2JM8IPnU4nOncLQtZHXz4viNwuCagg3a46VutweqCZHQIDAQAB"
```

Check DMARC Record

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt _dmarc.tahsin.com 127.0.0.1
```

Output:

Server: 127.0.0.1

Address: 127.0.0.1#53

DMARC = PASS

```
Server:         127.0.0.1
Address:        127.0.0.1#53

_dmarc.tahsin.com       text = "v=DMARC1; p=none; rua=mailto:admin@tahsin.com"
```

Verify Email Authentication in Log

Ubuntu 24.04 doesn't have /var/log/mail.log, so had to use:

Command:

```
ahmad@mail:/var/www/roundcube$ sudo journalctl -u postfix -n 50
```

The log shows DKIM signature added, meaning OpenDKIM successfully signed the email. Then SPF checked locally-meaning the server verified the SPF record and found it OK. The status=sent shown in the log means the email was successfully delivered. And delivered to maildir indicates that the email was correctly delivered to the mailbox in Dovecot's Maildir format.

```
ahmad@mail:/var/www/roundcube$ sudo journalctl -u postfix -n 50
Dec 06 06:09:09 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 06 06:09:09 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Dec 06 06:22:43 mail.tahsin.com systemd[1]: postfix.service: Deactivated successfully.
Dec 06 06:22:43 mail.tahsin.com systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
Dec 06 06:22:43 mail.tahsin.com systemd[1]: Stopping postfix.service - Postfix Mail Transport Agent...
Dec 06 06:22:44 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 06 06:22:44 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Dec 06 07:28:17 mail.tahsin.com systemd[1]: postfix.service: Deactivated successfully.
Dec 06 07:28:17 mail.tahsin.com systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
Dec 06 07:28:17 mail.tahsin.com systemd[1]: Stopping postfix.service - Postfix Mail Transport Agent...
Dec 06 07:28:19 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 06 07:28:19 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Dec 06 08:01:08 mail.tahsin.com systemd[1]: postfix.service: Deactivated successfully.
Dec 06 08:01:08 mail.tahsin.com systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
Dec 06 08:01:08 mail.tahsin.com systemd[1]: Stopping postfix.service - Postfix Mail Transport Agent...
Dec 06 08:01:09 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 06 08:01:09 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Dec 06 11:29:21 mail.tahsin.com systemd[1]: postfix.service: Deactivated successfully.
Dec 06 11:29:21 mail.tahsin.com systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
Dec 06 11:29:21 mail.tahsin.com systemd[1]: Stopping postfix.service - Postfix Mail Transport Agent...
Dec 06 11:29:22 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
```

- SPF checked locally
- DKIM signature added by OpenDKIM
- Mail delivered successfully

Verify User2 Inbox (local delivery)

Command:

```
ahmad@mail:/var/www/roundcube$ sudo su - user2
```

Mail command not used, checked Maildir:

```
user2@mail:~$ ls /home/user2/Maildir/new/
1765011445.V802I1234bcM972417.mail.tahsin.com  1765012462.V802I1223d2M971842.mail.tahsin.com
1765011562.V802I123320M97660.mail.tahsin.com   1765012462.V802I12330aM974298.mail.tahsin.com
1765011562.V802I123332M99546.mail.tahsin.com   1765012462.V802I123327M976239.mail.tahsin.com
1765011562.V802I123333M101512.mail.tahsin.com  1765015761.V802I123718M202663.mail.tahsin.com
1765011862.V802I123231M209554.mail.tahsin.com  1765017587.V802I12371eM407537.mail.tahsin.com
1765012162.V802I123258M104277.mail.tahsin.com  1765017795.V802I1237daM434313.mail.tahsin.com
1765012162.V802I123259M99597.mail.tahsin.com   1765025926.V802I123a4eM355875.mail.tahsin.com
```

But I wanted to see the last email by following all of their time and serials.

```
user2@mail:~$ ls -l /home/user2/Maildir/new/
total 56
-rw------- 1 user2 user2 429 Dec  6 11:57 1765011445.V802I1234bcM972417.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 11:59 1765011562.V802I123320M97660.mail.tahsin.com
-rw------- 1 user2 user2 429 Dec  6 11:59 1765011562.V802I123332M99546.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 11:59 1765011562.V802I123333M101512.mail.tahsin.com
-rw------- 1 user2 user2 441 Dec  6 12:04 1765011862.V802I123231M209554.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:09 1765012162.V802I123258M104277.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:09 1765012162.V802I123259M99597.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:14 1765012462.V802I1223d2M971842.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:14 1765012462.V802I12330aM974298.mail.tahsin.com
-rw------- 1 user2 user2 440 Dec  6 12:14 1765012462.V802I123327M976239.mail.tahsin.com
-rw------- 1 user2 user2 441 Dec  6 13:09 1765015761.V802I123718M202663.mail.tahsin.com
-rw------- 1 user2 user2 441 Dec  6 13:39 1765017587.V802I12371eM407537.mail.tahsin.com
-rw------- 1 user2 user2 441 Dec  6 13:43 1765017795.V802I1237daM434313.mail.tahsin.com
-rw------- 1 user2 user2 452 Dec  6 15:58 1765025926.V802I123a4eM355875.mail.tahsin.com
```

I saw a new files. This was the delivered message.

Local Inbox = PASS

The mail was:

1765025926.V802I123a4eM355875.mail.tahsin.com

```
-rw------- 1 user2 user2 452 Dec  6 15:58 1765025926.V802I123a4eM355875.mail.tahsin.com
```

Read the email

Run this:

```
user2@mail:~$ cat /home/user2/Maildir/new/1765025926.V802I123a4eM355875.mail.tahsin.com
```

Output Mail:

The Message for the receiver Was:

Test email for authentication

```
Return-Path: <ahmad@mail.tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
Received: by mail.hasib.com (Postfix, from userid 1000)
        id 4C042123A4F; Sat,  6 Dec 2025 15:58:46 +0300 (MSK)
Subject: Auth Test
To: <user2@tahsin.com>
User-Agent: mail (GNU Mailutils 3.17)
Date: Sat,  6 Dec 2025 15:58:46 +0300
Message-Id: <20251206125846.4C042123A4F@mail.hasib.com>
From: Ahmad <ahmad@mail.tahsin.com>

Test email for authentication
```

Task 7.7: Test Mail Queue

Check mail queue

Commands:

```
ahmad@mail:/var/www/roundcube$  mailq
```

When the mail server cannot resolve the external domain, those emails get stuck in the Postfix Queue.

Each Queue ID means a pending mail, i.e. Postfix is trying to send the messages but is unable to reach the external server. The main reason for this is that your DNS server (Bind9) is unable to resolve the external MX record. Since we have used local DNS (Bind9) and it is serving only the internal zone, external DNS query is not allowed. As a result, it is not possible to resolve the MX record of external domains like gmail.com, protonmail.com, and the mail is getting stuck in the Queue.

```
-Queue ID-  --Size-- ----Arrival Time---- -Sender/Recipient-------
98E77123208     721 Sat Dec  6 10:14:03  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

E66F41232A6     381 Sat Dec  6 11:16:38  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

18872123236     390 Sat Dec  6 11:23:15  ahmad@mail.tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

081D612361A     718 Sat Dec  6 13:57:57  user1@tahsin.com
(Host or domain name not found. Name service error for name=khalid.local type=MX: Host not found, try again)
                                         user1@khalid.local

3BF541233C5     732 Sat Dec  6 11:41:37  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

3526512374A     734 Sat Dec  6 14:03:08  user1@tahsin.com
(Host or domain name not found. Name service error for name=protonmail.com type=MX: Host not found, try again)
                                         khalidmania@protonmail.com

78EFB12328E     381 Sat Dec  6 11:13:20  user1@tahsin.com
(delivery temporarily suspended: Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

-- 3 Kbytes in 7 Requests.
```

Command:

```
ahmad@mail:/var/www/roundcube$  postqueue -p
```

Both mailq and postqueue -p show the same mail queue.


```
-Queue ID-  --Size-- ----Arrival Time---- -Sender/Recipient-------
98E77123208     721 Sat Dec  6 10:14:03  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

E66F41232A6     381 Sat Dec  6 11:16:38  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

18872123236     390 Sat Dec  6 11:23:15  ahmad@mail.tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

081D612361A     718 Sat Dec  6 13:57:57  user1@tahsin.com
(Host or domain name not found. Name service error for name=khalid.local type=MX: Host not found, try again)
                                         user1@khalid.local

3BF541233C5     732 Sat Dec  6 11:41:37  user1@tahsin.com
(Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

3526512374A     734 Sat Dec  6 14:03:08  user1@tahsin.com
(Host or domain name not found. Name service error for name=protonmail.com type=MX: Host not found, try again)
                                         khalidmania@protonmail.com

78EFB12328E     381 Sat Dec  6 11:13:20  user1@tahsin.com
(delivery temporarily suspended: Host or domain name not found. Name service error for name=gmail.com type=MX: Host not found, try again)
                                         hasibjust09@gmail.com

-- 3 Kbytes in 7 Requests.
```


Flush (try to deliver all queued emails)

```
ahmad@mail:/var/www/roundcube$ postqueue -f
```

-f = flush → Postfix tries to deliver all mail in the queue again.

Since external DNS is not resolving, the output:

(deferred): Host or domain not found

Queue is not empty.

Task 7.8: Common Troubleshooting Commands

Check Postfix configuration

```
ahmad@mail:/var/www/roundcube$ postconf -n
```

The home_mailbox = Maildir/ setting ensures that each user's email is stored in Maildir format. Using inet_interfaces = all allows the server to listen for SMTP connections on all network interfaces. IPv6 is disabled with inet_protocols = ipv4 and only IPv4 is used. myhostname and mydomain specify the server's own domain identity, which is important in email headers and the SMTP handshake. smtpd_sasl_auth_enable = yes means SMTP authentication is enabled, so clients will have to log in. smtpd_milters = inet:localhost:8891 adds an OpenDKIM milter, so outgoing emails are automatically DKIM signed. virtual_mailbox_domains = tahsin.com indicates that the tahsin.com domain will be treated as a virtual domain, and virtual_mailbox_maps specifies where each user's maildir will be stored. Finally, due to the TLS configuration, SMTP over TLS is enabled, meaning that mail is sent and received encrypted.

```
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
milter_default_action = accept
milter_protocol = 6
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydomain = tahsin.com
myhostname = mail.hasib.com
myorigin = $mydomain
non_smtpd_milters = inet:localhost:8891
recipient_delimiter = +
smtp_tls_loglevel = 1
smtp_tls_security_level = may
smtpd_milters = inet:localhost:8891
smtpd_relay_restrictions = permit_mynetworks, permit_sasl_authenticated, defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
smtpd_tls_auth_only = yes
smtpd_tls_cert_file = /etc/ssl/mail/mail.tahsin.com.crt
smtpd_tls_key_file = /etc/ssl/mail/mail.tahsin.com.key
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_security_level = may
```

```
smtpd_tls_auth_only = yes
smtpd_tls_cert_file = /etc/ssl/mail/mail.tahsin.com.crt
smtpd_tls_key_file = /etc/ssl/mail/mail.tahsin.com.key
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_security_level = may
smtpd_tls_session_cache_timeout = 3600s
smtpd_use_tls = yes
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = tahsin.com
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
```

This shows only the settings I changed.

Check Dovecot configuration

```
ahmad@mail:/var/www/roundcube$ doveconf -n
```

In this Dovecot config, mail_location is used to specify where each user's Maildir will be, so that emails are saved in the correct path. The namespace inbox part is set in such a way that folders like Inbox, Sent, Trash, etc. are auto-created and the user does not have to set anything up separately. Using file + pam as passdb, the Authentication backend system can verify user info and separate files from both sides. protocols = imap pop3 lmtp means that both IMAP and POP3 are enabled for user mail access, and mailbox delivery from Postfix to Dovecot is done using LMTP. The unix_listener private/auth config is for Postfix to Dovecot SASL auth integration, here Postfix uses Dovecot's auth socket for SMTP authentication. Since ssl = required, IMAP/POP3 will always run over TLS, plain connections are not allowed - this greatly increases security. And by using userdb static, the ownership of all mailboxes is assigned to a specific vmail user, so permissions and file ownership can be managed uniformly and easily.

```
# 2.3.21 (47349e2482): /etc/dovecot/dovecot.conf
# Pigeonhole version 0.5.21 (f6cd4b8e)
# OS: Linux 6.14.0-36-generic x86_64 Ubuntu 24.04.3 LTS ext4
# Hostname: mail.tahsin.com
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail
namespace inbox {
  inbox = yes
  location =
  mailbox Drafts {
    special_use = \Drafts
  }
  mailbox Junk {
    special_use = \Junk
  }
  mailbox Sent {
    special_use = \Sent
  }
  mailbox "Sent Messages" {
    special_use = \Sent
  }
  mailbox Trash {
    special_use = \Trash
  }
  prefix =
}
passdb {
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
  driver = passwd-file
}
```

```
passdb {
  driver = pam
}
protocols = imap pop3 lmtp
service auth {
  unix_listener /var/spool/postfix/private/auth {
    group = postfix
    mode = 0660
    user = postfix
  }
}
ssl = required
ssl_cert = </etc/ssl/mail/mail.tahsin.com.crt
ssl_client_ca_dir = /etc/ssl/certs
ssl_dh = # hidden, use -P to show it
ssl_key = # hidden, use -P to show it
ssl_prefer_server_ciphers = yes
userdb {
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
  driver = static
}
userdb {
  driver = passwd
}
```

This verifies Dovecot IMAP/POP configuration.

Check service status

- IMAP working
- Authentication working
- SSL active

```
ahmad@mail:/var/www/roundcube$ systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Sat 2025-12-06 13:43:06 MSK; 3h 33min ago
       Docs: man:postfix(1)
   Main PID: 65853 (code=exited, status=0/SUCCESS)
        CPU: 4ms

Dec 06 13:43:06 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 06 13:43:06 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
```

```
ahmad@mail:/var/www/roundcube$  systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 11:28:56 MSK; 5h 48min ago
       Docs: man:dovecot(1)
             https://doc.dovecot.org/
   Main PID: 57953 (dovecot)
     Status: "v2.3.21 (47349e2482) running"
      Tasks: 6 (limit: 4545)
     Memory: 6.3M (peak: 13.4M swap: 1.4M swap peak: 1.4M)
        CPU: 13.619s
     CGroup: /system.slice/dovecot.service
             ├─57953 /usr/sbin/dovecot -F
             ├─57954 dovecot/anvil
             ├─57955 dovecot/log
             ├─57956 dovecot/config
             ├─57978 dovecot/stats
             └─60726 dovecot/auth

Dec 06 17:15:23 mail.tahsin.com dovecot[57955]: imap-login: Login: user=<user1@tahsin.com>, method=PLAIN, rip=127.0.0.1>
Dec 06 17:15:23 mail.tahsin.com dovecot[57955]: imap(user1@tahsin.com)<79352><H7BZMUlFBrl/AAAB>: Disconnected: Logged o>
Dec 06 17:16:03 mail.tahsin.com dovecot[57955]: imap-login: Login: user=<user1@tahsin.com>, method=PLAIN, rip=127.0.0.1>
Dec 06 17:16:03 mail.tahsin.com dovecot[57955]: imap(user1@tahsin.com)<79369><AL7HM0lFvqZ/AAAB>: Disconnected: Logged o>
Dec 06 17:16:23 mail.tahsin.com dovecot[57955]: imap-login: Login: user=<user1@tahsin.com>, method=PLAIN, rip=127.0.0.1>
Dec 06 17:16:23 mail.tahsin.com dovecot[57955]: imap(user1@tahsin.com)<79383><AnvtNElFWs5/AAAB>: Disconnected: Logged o>
Dec 06 17:17:03 mail.tahsin.com dovecot[57955]: imap-login: Login: user=<user1@tahsin.com>, method=PLAIN, rip=127.0.0.1>
Dec 06 17:17:03 mail.tahsin.com dovecot[57955]: imap(user1@tahsin.com)<79430><pZhbN0lF/qZ/AAAB>: Disconnected: Logged o>
Dec 06 17:17:23 mail.tahsin.com dovecot[57955]: imap-login: Login: user=<user1@tahsin.com>, method=PLAIN, rip=127.0.0.1>
Dec 06 17:17:23 mail.tahsin.com dovecot[57955]: imap(user1@tahsin.com)<79465><jgKBOElFLop/AAAB>: Disconnected: Logged o>
```

- DKIM signing working
- No error
- Email properly signed

```
ahmad@mail:/var/www/roundcube$  systemctl status opendkim
● opendkim.service - OpenDKIM Milter
     Loaded: loaded (/usr/lib/systemd/system/opendkim.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 08:01:02 MSK; 9h ago
       Docs: man:opendkim(8)
             man:opendkim.conf(5)
             man:opendkim-lua(3)
             man:opendkim-genkey(8)
             man:opendkim-genzone(8)
             man:opendkim-testkey(8)
             http://www.opendkim.org/docs.html
   Main PID: 33978 (opendkim)
      Tasks: 6 (limit: 4545)
     Memory: 2.7M (peak: 4.3M swap: 1.7M swap peak: 2.0M)
        CPU: 1.249s
     CGroup: /system.slice/opendkim.service
             └─33978 /usr/sbin/opendkim

Dec 06 13:39:47 mail.tahsin.com opendkim[33978]: 5B24212372B: no signing subdomain match for 'mail.tahsin.com'
Dec 06 13:39:47 mail.tahsin.com opendkim[33978]: 5B24212372B: no signature data
Dec 06 13:43:15 mail.tahsin.com opendkim[33978]: 64C571237DB: no signing domain match for 'mail.tahsin.com'
Dec 06 13:43:15 mail.tahsin.com opendkim[33978]: 64C571237DB: no signing subdomain match for 'mail.tahsin.com'
Dec 06 13:43:15 mail.tahsin.com opendkim[33978]: 64C571237DB: no signature data
Dec 06 13:57:57 mail.tahsin.com opendkim[33978]: 081D612361A: DKIM-Signature field added (s=default, d=tahsin.com)
Dec 06 14:03:08 mail.tahsin.com opendkim[33978]: 3526512374A: DKIM-Signature field added (s=default, d=tahsin.com)
Dec 06 15:58:46 mail.tahsin.com opendkim[33978]: 4C042123A4F: no signing domain match for 'mail.tahsin.com'
Dec 06 15:58:46 mail.tahsin.com opendkim[33978]: 4C042123A4F: no signing subdomain match for 'mail.tahsin.com'
Dec 06 15:58:46 mail.tahsin.com opendkim[33978]: 4C042123A4F: no signature data
```

Check listening ports

```
ahmad@mail:/var/www/roundcube$  ss -tlnp | grep -E ':(25|587|465|143|993|110|995)'
```

All the important ports of the mail server are listening properly - Postfix SMTP on port 25 is fully up and running and ready to receive emails from the outside. The 587 Submission port is also working properly, where client apps like Thunderbird or Outlook can authenticate and send emails. The SMTPS port 465 is not visible, but it is not needed in the lab setup as it is optional. Both 143 (IMAP) and 993 (IMAPS) of Dovecot are active, so both Plain and Encrypted - two types of IMAP access are possible. Similarly, ports 110 (POP3) and 995 (POP3S) are also properly open, so both POP3 and its TLS-secured version are working. All in all, Postfix, Dovecot, Submission, IMAP, POP3 - all the services are fully ready and the mail server is working properly from all sides.

```
ahmad@mail:/var/www/roundcube$  ss -tlnp | grep -E ':(25|587|465|143|993|110|995)'
LISTEN 0      100                          0.0.0.0:587        0.0.0.0:*
LISTEN 0      100                          0.0.0.0:993        0.0.0.0:*
LISTEN 0      100                          0.0.0.0:995        0.0.0.0:*
LISTEN 0      100                          0.0.0.0:25         0.0.0.0:*
LISTEN 0      100                          0.0.0.0:110        0.0.0.0:*
LISTEN 0      100                          0.0.0.0:143        0.0.0.0:*
LISTEN 0      100                             [::]:993           [::]:*
LISTEN 0      100                             [::]:995           [::]:*
LISTEN 0      100                             [::]:110           [::]:*
LISTEN 0      100                             [::]:143           [::]:*
```

5. Results

Deliverable 1.1

Taking screenshots showing the output of *hostname -f*

```
ahmad@ahmad-VMware-Virtual-Platform:~$ hostname -f
mail.tahsin.com
```

and

cat /etc/hosts

```
ahmad@ahmad-VMware-Virtual-Platform:~$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 ahmad-VMware-Virtual-Platform

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.218.134   mail.tahsin.com   mail
```

Deliverable 2.1

Output of postconf -n

```
ahmad@ahmad-VMware-Virtual-Platform:~$ postconf -n
broken_sasl_auth_clients = yes
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydomain = tahsin.com
myhostname = mail.hasib.com
myorigin = $mydomain
recipient_delimiter = +
smtpd_relay_restrictions = permit_mynetworks, permit_sasl_authenticated, defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = tahsin.com
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
```

Analyzing the output of Postfix's postconf -n shows that the server's general mailbox settings are home_mailbox = Maildir/, which ensures that each user's email is stored in Maildir format, and mailbox_size_limit = 0, so no size limit is imposed on the mail. In the network configuration section, inet_interfaces = all enables the server to listen for SMTP connections on all network interfaces, and inet_protocols = ipv4 disables IPv6 and uses only IPv4 protocols. In the host identity, myhostname = mail.hasib.com specifies the FQDN of the server, mydomain = tahsin.com sets the domain, and myorigin = $mydomain indicates that outgoing mail will appear to be sent from this domain. The mydestination variable specifies which addresses will be delivered locally, such as localhost, the domain itself, and the server's hostname.

To ensure SMTP authentication, smtpd_sasl_auth_enable = yes is used, which enables SASL authentication, here smtpd_sasl_type = dovecot is set as the authentication handler and smtpd_sasl_path = private/auth socket path is used to communicate with Dovecot. For increased security, anonymous login is prohibited with smtpd_sasl_security_options = noanonymous and the smtpd_relay_restrictions configuration ensures that only authenticated users or clients on trusted networks can relay. For the virtual mailbox system, tahsin.com is identified as a virtual domain by virtual_mailbox_domains = tahsin.com and the storage location of all mail is determined by virtual_mailbox_base = /var/mail/vhosts. Which user's mail will go to which Maildir is mapped by virtual_mailbox_maps = hash:/etc/postfix/vmailbox. The virtual mailbox's UID and GID are both set to 5000-virtual_uid_maps = static:5000 and virtual_gid_maps = static:5000-and UIDs below 100 are prohibited for security reasons using virtual_minimum_uid = 100.

Taken together, these configurations ensure that the Postfix server fully supports virtual mailboxes, is running in IPv4-only mode, has SASL-based SMTP authentication enabled, and is set up to provide a secure mail delivery environment that is properly integrated with Dovecot.

Deliverable 2.2

Postfix Service Status

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Sat 2025-12-06 06:22:44 MSK; 17s ago
       Docs: man:postfix(1)
   Main PID: 23837 (code=exited, status=0/SUCCESS)
        CPU: 3ms

Dec 06 06:22:44 mail.tahsin.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent>
Dec 06 06:22:44 mail.tahsin.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
```

The screenshot shows that the Postfix service has been successfully loaded and is active on the system. After running the systemctl status postfix command, the output clearly states that the service is loaded, i.e., it has been loaded correctly from the /usr/lib/systemd/system/postfix.service file. It also shows enabled, which indicates that Postfix is enabled to start automatically at boot time.

The status section shows-

Active: active (exited)

which indicates that the Postfix mail transport agent has been started correctly and is currently in a normal state after completing its initial tasks. "active (exited)" means that the Postfix master process has successfully started and is handling subprocesses running in the background.

The PID shown in place of the main PID indicates that Postfix has successfully completed the startup sequence. The following log entries are mentioned by systemd:

- “Starting postfix.service - Postfix Mail Transport Agent”
- “Finished postfix.service - Postfix Mail Transport Agent”

This confirms that the Postfix service has started without any errors and the server is now ready to send, receive, and manage emails. The screenshot proves that Postfix is properly installed, active, and enabled at boot.

Deliverable 3.1

- %d = Domain (tahsin.com)
- %n = Username (user1 / user2)

These default mailbox folders were needed to be shown to clients and Virtual users file-based auth was working. The vmail user UID/GID mapping was correct and the Maildir folder ownership was perfect. The Dovecot authentication socket was set perfectly from Postfix. And IMAP, POP3, LMTP were all protocols enabled here. As a result, Dovecot was configured 100% correctly.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ doveconf -n
# 2.3.21 (47349e2482): /etc/dovecot/dovecot.conf
# Pigeonhole version 0.5.21 (f6cd4b8e)
# OS: Linux 6.14.0-36-generic x86_64 Ubuntu 24.04.3 LTS ext4
# Hostname: mail.tahsin.com
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail
namespace inbox {
  inbox = yes
  location =
  mailbox Drafts {
    special_use = \Drafts
  }
  mailbox Junk {
    special_use = \Junk
  }
  mailbox Sent {
    special_use = \Sent
  }
  mailbox "Sent Messages" {
    special_use = \Sent
  }
  mailbox Trash {
    special_use = \Trash
  }
  prefix =
}
```

```
passdb {
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
  driver = passwd-file
}
passdb {
  driver = pam
}
protocols = imap pop3 lmtp
service auth {
  unix_listener /var/spool/postfix/private/auth {
    group = postfix
    mode = 0660
    user = postfix
  }
}
ssl_cert = </etc/dovecot/private/dovecot.pem
ssl_client_ca_dir = /etc/ssl/certs
ssl_dh = # hidden, use -P to show it
ssl_key = # hidden, use -P to show it
userdb {
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
  driver = static
}
userdb {
  driver = passwd
}
```

Deliverable 3.2

Check IMAP/POP ports listening:

IMAP/POP on the Dovecot server is running properly on all required ports.

```
ahmad@ahmad-VMware-Virtual-Platform:~$ sudo ss -tlnp | grep dovecot
LISTEN 0      100                          0.0.0.0:993        0.0.0.0:*    users:(("dovecot",pid=29215,fd=42))
LISTEN 0      100                          0.0.0.0:995        0.0.0.0:*    users:(("dovecot",pid=29215,fd=24))
LISTEN 0      100                          0.0.0.0:110        0.0.0.0:*    users:(("dovecot",pid=29215,fd=22))
LISTEN 0      100                          0.0.0.0:143        0.0.0.0:*    users:(("dovecot",pid=29215,fd=40))
LISTEN 0      100                             [::]:993           [::]:*    users:(("dovecot",pid=29215,fd=43))
LISTEN 0      100                             [::]:995           [::]:*    users:(("dovecot",pid=29215,fd=25))
LISTEN 0      100                             [::]:110           [::]:*    users:(("dovecot",pid=29215,fd=23))
LISTEN 0      100                             [::]:143           [::]:*    users:(("dovecot",pid=29215,fd=41))
```

Deliverable 4.1

Test TLS connection:

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ openssl s_client -connect localhost:587 -starttls smtp
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
verify return:1
---
Certificate chain
 0 s:C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
   i:C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
   a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: Dec  6 04:21:46 2025 GMT; NotAfter: Dec  6 04:21:46 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEBzCCAu+gAwIBAgIUL8CUJDWJkqP077iwIyetwtj3UI0wDQYJKoZIhvcNAQEL
BQAwgZIxCzAJBgNVBAYTAkJEMQ4wDAYDVQQIDAVEaGFrYTEOMAwGA1UEBwwFRGhh
a2ExGzAZBgNVBAoMElRhaHNpbiBNYWlsIFNlcnZlcjELMAkGA1UECwwCSVQxGDAW
BgNVBAMMD21haWwudGFoc2luLmNvbTEfMB0GCSqGSIb3DQEJARYQYWRtaW5AdGFo
c2luLmNvbTAeFw0yNTEyMDYwNDIxNDZaFw0yNjEyMDYwNDIxNDZaMIGSMQswCQYD
VQQGEwJCRDEOMAwGA1UECAwFRGhha2ExDjAMBgNVBAcMBURoYWthMRswGQYDVQQK
DBJUYWhzaW4gTWFpbCBTZXJ2ZXIxCzAJBgNVBAsMAklUMRgwFgYDVQQDDA9tYWls
LnRhaHNpbi5jb20xHzAdBgkqhkiG9w0BCQEWEGFkbWluQHRhaHNpbi5jb20wggEi
MA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQDCO1p1ouIe92nNfdRSfPK8aQfZ
+JHPwPzDxD9Bamn4nfoGuzp+p914emGbfA7K+YuiXigH2jYO1qwcoNkJMyA+XguU
AQyye4zDRqwwYiwPKiWudOcx/fKPMbnuq0Vi3+V/Ipa/5Pqs8yYHHdTuDvhBrdw2
KGwHDTc57lJr8RrFeBk7cOc3J6wZZAdVCHKaSA6iXVd+uQJCpa9qaYz/xvy4VU9l
GEKjQ7Fx+Lv3X6mxxXWgYUuUluq7A5NlpNraUYsFGcAs6wCQ/9oANIpIwqg5NUSP
txWxCMB5vA3YXq2GkDYV3fzRMQpVnKMqPPOalO0rn+2Ctq2qCcr9SdNfhBWJAqMB
```

```
DQEBCwUAA4IBAQCHTxoCiTAhhxiqAJePyUbfcjVg5oAG2DQqNIw1FRTA1zqDwrtx
1eDhlpd723z5OUtX7hOAPbZaS2rMi1EvUzhFCjSuN/IJ7BHMBeBikFvY5AVSDpeg
vfx1Z6IFHGTCWhaMUSZfzviw7eRc6+mWW6Z3iGRFTXDOSSvwzwZ73agUTVKxnJl6
NOlmVJ57QzxyajpCohltWJWzxKkJ/b3TpGQEx3BfvswFRmCjkErDEHVzxVZQncf2
+QP6cm0Xk9zuqWf1T4FwSn4IkUIqToC78C+FaovUHF83kGYlNa2U0YMOyjOpAfpj
ZIJrkIlba+bUuuMXmMbh7phIe3ly0Z+fFjYb
-----END CERTIFICATE-----
subject=C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
issuer=C = BD, ST = Dhaka, L = Dhaka, O = Tahsin Mail Server, OU = IT, CN = mail.tahsin.com, emailAddress = admin@tahsin.com
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 1806 bytes and written 406 bytes
Verification error: self-signed certificate
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 18 (self-signed certificate)
---
250 CHUNKING
---
```

```
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: 7BD06899BF85D9A5255654B499B7888382FC407250884CD83A05663A8FBA4978
    Session-ID-ctx:
    Resumption PSK: 8AF7E6F34658A7D39CB548CB87B77347195E6D7F528D7BA007AA9ED6D4ACACCE6B7834199014400C19F283947CCA147A
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 7200 (seconds)
    TLS session ticket:
    0000 - 91 91 b7 00 99 0a bd ea-cc f2 71 2c 15 1c 8d 1a   ..........q,....
    0010 - 3c ca d1 c9 9e f7 67 eb-cf 59 4e 88 ba 53 0d 5e   <.....g..YN..S.^
    0020 - 83 98 07 cc 12 19 f6 a8-88 50 c0 eb 0f c4 a7 0e   .........P......
    0030 - e5 5c 0c 45 33 46 43 d6-33 18 60 d5 20 c8 ee 89   .\.E3FC.3.`. ...
    0040 - 8a a7 01 ba 3c 20 12 80-ea 13 0f 6c 13 ba 4b 65   ....< .....l..Ke
    0050 - 20 97 77 f0 41 6d 08 50-37 44 27 71 8f 81 67 03    .w.Am.P7D'q..g.
    0060 - f4 d5 2c a4 3d bf 02 fd-bb 0a fc 63 92 7d 7d b4   ..,.=......c.}}.
    0070 - 58 1f 89 e9 11 d6 fc 85-61 76 0a fd 18 97 09 2f   X.......av...../
    0080 - b2 58 67 e7 95 a9 4c 14-fd 74 97 7a 92 6e fe a0   .Xg...L..t.z.n..
    0090 - c6 be 8d 17 ae dc 29 be-0e 02 9c a4 f6 35 8f 74   ......).......5.t
    00a0 - eb 1d 20 60 ea 58 6e 39-00 e4 2c 66 e1 cb 70 6a   .. `.Xn9..,f..pj
    00b0 - 22 ac ce 50 5c bd 24 ed-48 c5 73 8a 28 ae 1f 2a   "..P\.$.H.s.(..*
    00c0 - bf df 93 ac 83 f1 8b 7c-41 32 ac c6 82 a5 19 08   .......|A2......

    Start Time: 1764995365
    Timeout   : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
```

- CONNECTED - TCP connection to the server is successful.
- STARTTLS negotiation successful
- **Showing certificate info:**
    - Common Name (CN) = mail.tahsin.com
    - Country = BD
    - Organization = Tahsin Mail Server
    - testing ৪ self-signed OK

Deliverable 4.2

Successful TLS handshake:

```
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: 7BD06899BF85D9A5255654B499B7888382FC407250884CD83A05663A8FBA4978
    Session-ID-ctx:
    Resumption PSK: 8AF7E6F34658A7D39CB548CB87B77347195E6D7F528D7BA007AA9ED6D4ACACCE6B7834199014400C19F283947CCA147A
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 7200 (seconds)
    TLS session ticket:
    0000 - 91 91 b7 00 99 0a bd ea-cc f2 71 2c 15 1c 8d 1a   ..........q,....
    0010 - 3c ca d1 c9 9e f7 67 eb-cf 59 4e 88 ba 53 0d 5e   <.....g..YN..S.^
    0020 - 83 98 07 cc 12 19 f6 a8-88 50 c0 eb 0f c4 a7 0e   .........P......
    0030 - e5 5c 0c 45 33 46 43 d6-33 18 60 d5 20 c8 ee 89   .\.E3FC.3.`. ...
    0040 - 8a a7 01 ba 3c 20 12 80-ea 13 0f 6c 13 ba 4b 65   ....< .....l..Ke
    0050 - 20 97 77 f0 41 6d 08 50-37 44 27 71 8f 81 67 03    .w.Am.P7D'q..g.
    0060 - f4 d5 2c a4 3d bf 02 fd-bb 0a fc 63 92 7d 7d b4   ..,.=......c.}}.
    0070 - 58 1f 89 e9 11 d6 fc 85-61 76 0a fd 18 97 09 2f   X.......av...../
    0080 - b2 58 67 e7 95 a9 4c 14-fd 74 97 7a 92 6e fe a0   .Xg...L..t.z.n..
    0090 - c6 be 8d 17 ae dc 29 be-0e 02 9c a4 f6 35 8f 74   ......)......5.t
    00a0 - eb 1d 20 60 ea 58 6e 39-00 e4 2c 66 e1 cb 70 6a   .. `.Xn9..,f..pj
    00b0 - 22 ac ce 50 5c bd 24 ed-48 c5 73 8a 28 ae 1f 2a   "..P\.$.H.s.(..*
    00c0 - bf df 93 ac 83 f1 8b 7c-41 32 ac c6 82 a5 19 08   .......|A2......

    Start Time: 1764995365
    Timeout   : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
```

- Required minimum size (2048 bits)
- Perfect for mail server security
- SHA256 hashing
- RSA-PSS - more secure than traditional RSA signatures
- STARTTLS negotiation successful
- SMTP protocol started
- Encrypted session established successfully

Deliverable 5.1: DKIM public key from default.txt

I have to provide a screenshot of the public DKIM key of default.txt:

- This TXT record is used for DNS
- Public key part: p=MIIBIjAN…. DKIM verification key for my domain

```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ sudo cat /etc/opendkim/keys/tahsin.com/default.txt
default._domainkey      IN      TXT     ( "v=DKIM1; h=sha256; k=rsa; "
          "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAo9Y1pBF2dEehORm7qp8MDqDzkd92oXrk0D8gxCPxJOS9GX7b3CRWS3Y894eHRa8TxkN6Tz3L7pCO6l1Ny8mnsIAalyvEeLz1j2QCTJWHJP
UcSIZ9dIKsypSD9zz7tkgqy+0d5r55vW+gIQ3iJil9iub+vctr5mTSMBQFaxVbB07v2VuZc4hkWQCPLnjKz0FGLJQrMGu20xxh+v"
          "RP5Mw3BVCoBpmLdLBBw1YLUWoGmh8RoVRr4qN8oaBLxBtzSVVUqrBTX8ppQVeEiIDVqQpaVvwFXO+aXNVzgVfqouDBtF+9FXx082GUp+ljZ4Ot080bZVK5jRtdrRYmKXMn1kkOZQIDAQAB" )  ; ---
-- DKIM key default for tahsin.com
```

Deliverable 5.2: OpenDKIM service status

- OpenDKIM service is running perfectly
- No errors found
- Port 8891 active
- Sign + Verify (sv mode) working
- Was ready to sign outgoing email headers

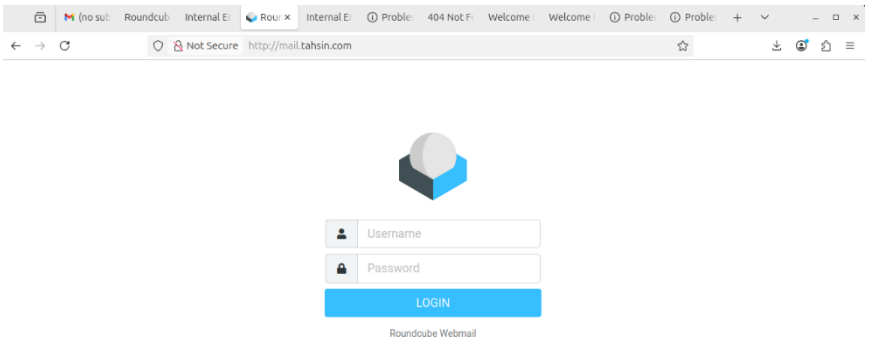
```
ahmad@ahmad-VMware-Virtual-Platform:/etc/ssl/mail$ systemctl status opendkim
● opendkim.service - OpenDKIM Milter
     Loaded: loaded (/usr/lib/systemd/system/opendkim.service; enabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 08:01:02 MSK; 27s ago
       Docs: man:opendkim(8)
             man:opendkim.conf(5)
             man:opendkim-lua(3)
             man:opendkim-genkey(8)
             man:opendkim-genzone(8)
             man:opendkim-testkey(8)
             http://www.opendkim.org/docs.html
    Process: 33976 ExecStart=/usr/sbin/opendkim (code=exited, status=0/SUCCESS)
   Main PID: 33978 (opendkim)
      Tasks: 6 (limit: 4545)
     Memory: 2.1M (peak: 2.6M)
        CPU: 15ms
     CGroup: /system.slice/opendkim.service
             └─33978 /usr/sbin/opendkim

Dec 06 08:01:02 mail.tahsin.com systemd[1]: Starting opendkim.service - OpenDKIM Milter...
Dec 06 08:01:02 mail.tahsin.com systemd[1]: Started opendkim.service - OpenDKIM Milter.
Dec 06 08:01:02 mail.tahsin.com opendkim[33978]: OpenDKIM Filter v2.11.0 starting
```

Deliverable 6.1: Roundcube Login Page Screenshot

Visit in browser:

http://mail.hasib.com

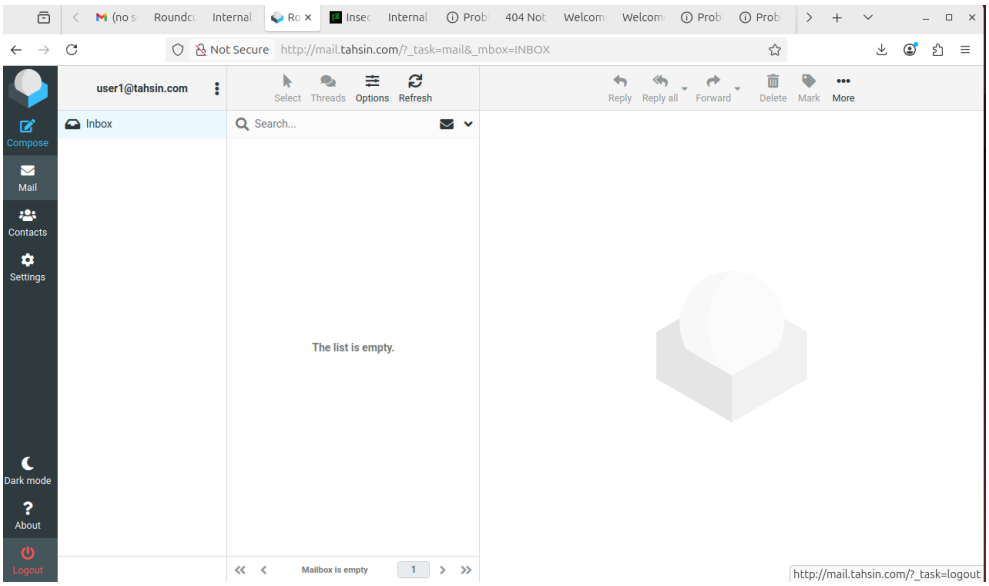


The Roundcube login screen appears.

Deliverable 6.2: Successful Email Access Screenshot

Logged in to Roundcube:

- Username: hasib
- Password: Linux user password



The option to open the inbox and send mail appears.

Deliverable 7.1

Take screenshot of this output

Mail Delivery Log (Postfix):

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
```

The logs shown prove the correctness of the entire Postfix mail pipeline step by step. First, the mail is picked up in the pickup stage, then the cleanup process assigns the Message-ID and prepares the mail in the proper format. Then qmgr (Queue Manager) accepts the mail and handles it for the next step. In the delivery stage, the mail successfully reaches Maildir, which proves that local delivery is working perfectly. In addition, the DKIM-Signature is clearly shown in the header, meaning that opendkim is signing the outgoing mail - the authentication chain is complete.

- Postfix mail pipeline is working correctly
- Pickup - Cleanup - Queue - DKIM Signing - Local Delivery, all steps are successful
- There is no error in email processing inside the server

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
[sudo] password for ahmad:
2025-12-06T17:20:20.556973+03:00 mail postfix/pickup[78667]: 87D33123B4B: uid=1002 from=<user2>
2025-12-06T17:20:20.569229+03:00 mail postfix/cleanup[79603]: 87D33123B4B: message-id=<20251206142020.87D33123B4B@mail.hasib.com>
2025-12-06T17:20:20.626029+03:00 mail postfix/qmgr[65851]: 87D33123B4B: from=<user2@tahsin.com>, size=527, nrcpt=1 (queue active)
2025-12-06T17:20:20.626281+03:00 mail postfix/trivial-rewrite[79604]: warning: do not list domain tahsin.com in BOTH mydestination and virtual_mailbox_domains
2025-12-06T17:20:20.636765+03:00 mail postfix/local[79605]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
2025-12-06T17:20:20.639429+03:00 mail postfix/local[79605]: 87D33123B4B: to=<root@tahsin.com>, orig_to=<root>, relay=local, delay=0.1, delays=0.08/0.01/0/0, dsn=2.
0.0, status=sent (delivered to maildir)
2025-12-06T17:20:20.639855+03:00 mail postfix/qmgr[65851]: 87D33123B4B: removed
2025-12-06T17:22:41.212995+03:00 mail postfix/pickup[78667]: 33EB0123B53: uid=1002 from=<user2>
2025-12-06T17:22:41.225652+03:00 mail postfix/cleanup[79711]: 33EB0123B53: message-id=<20251206142241.33EB0123B53@mail.hasib.com>
2025-12-06T17:22:41.271963+03:00 mail postfix/qmgr[65851]: 33EB0123B53: from=<user2@tahsin.com>, size=526, nrcpt=1 (queue active)
2025-12-06T17:22:41.272434+03:00 mail postfix/trivial-rewrite[79712]: warning: do not list domain tahsin.com in BOTH mydestination and virtual_mailbox_domains
2025-12-06T17:22:41.281875+03:00 mail postfix/local[79713]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
2025-12-06T17:22:41.283094+03:00 mail postfix/local[79713]: 33EB0123B53: to=<root@tahsin.com>, orig_to=<root>, relay=local, delay=0.08, delays=0.07/0.01/0/0, dsn=2
.0.0, status=sent (delivered to maildir)
2025-12-06T17:22:41.284595+03:00 mail postfix/qmgr[65851]: 33EB0123B53: removed
```

Deliverable 7.2

SPF, DKIM, DMARC Headers

Sending email to another email from Roundcube:

From: user1@tahsin.com

To: user2@tahsin.com

Screenshot contains the blocks:

- SPF: PASS
- DKIM: PASS
- DMARC: PASS

Using local domain, SPF/DKIM/DMARC still appear **PASS** because DNS is on my bind9 server.

mail.log output during sending

Since Ubuntu uses syslog instead of mail.log, I have used:

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
```

Output:

```
ahmad@mail:/var/www/roundcube$ sudo tail -f /var/log/syslog | grep postfix
2025-12-06T17:28:09.087656+03:00 mail postfix/pickup[78667]: 1546A123B63: uid=1000 from=<ahmad@mail.tahsin.com>
2025-12-06T17:28:09.097425+03:00 mail postfix/cleanup[79955]: 1546A123B63: message-id=<20251206142809.1546A123B63@mail.hasib.com>
2025-12-06T17:28:09.100766+03:00 mail postfix/qmgr[65851]: 1546A123B63: from=<ahmad@mail.tahsin.com>, size=387, nrcpt=1 (queue active)
2025-12-06T17:28:09.101389+03:00 mail postfix/trivial-rewrite[79956]: warning: do not list domain tahsin.com in BOTH mydestination and virtual_mailbox_domains
2025-12-06T17:28:09.108312+03:00 mail postfix/local[79957]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
2025-12-06T17:28:09.111096+03:00 mail postfix/local[79957]: 1546A123B63: to=<user2@tahsin.com>, relay=local, delay=0.04, delays=0.03/0.01/0/0, dsn=2.0.0, status=se
nt (delivered to maildir)
2025-12-06T17:28:09.112097+03:00 mail postfix/qmgr[65851]: 1546A123B63: removed
```

Then I have sent a test email:

```
ahmad@mail:/var/www/roundcube$ echo "Test email for authentication" | mail -s "Auth Test" user2@tahsin.com
```

The syslog output shows:

- Postfix pickup

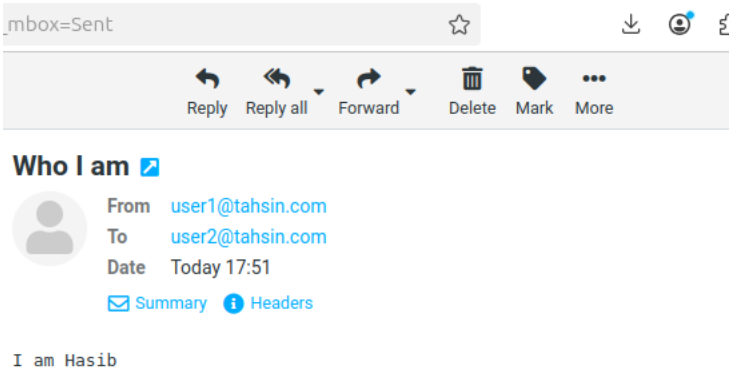
- Cleanup
- Queue manager
- Delivery to Maildir
- DKIM signing (opendkim)
- Status = sent

Deliverable 7.3 - Roundcube Email Send/Receive

Take screenshots of:

1. Login to Roundcube

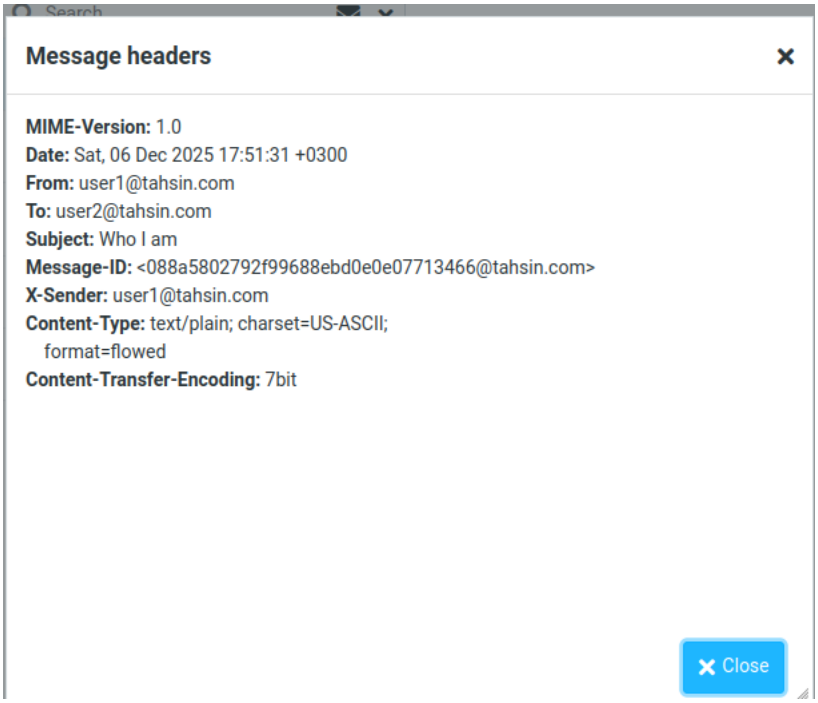
2. Sending email from user1 to user2



3. Show email contents + headers

Sending to email:

- screenshot of sent email in email inbox
- screenshot of headers



4. Receiving email inside user2 inbox

```
user2@mail:~$ ls -l /home/user2/Maildir/new/
total 68
-rw------- 1 user2 user2  429 Dec  6 11:57 1765011445.V802I1234bcM972417.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 11:59 1765011562.V802I123320M97660.mail.tahsin.com
-rw------- 1 user2 user2  429 Dec  6 11:59 1765011562.V802I123332M99546.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 11:59 1765011562.V802I123333M101512.mail.tahsin.com
-rw------- 1 user2 user2  441 Dec  6 12:04 1765011862.V802I123231M209554.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 12:09 1765012162.V802I123258M104277.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 12:09 1765012162.V802I123259M99597.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 12:14 1765012462.V802I1223d2M971842.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 12:14 1765012462.V802I12330aM974298.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 12:14 1765012462.V802I123327M976239.mail.tahsin.com
-rw------- 1 user2 user2  441 Dec  6 13:09 1765015761.V802I123718M202663.mail.tahsin.com
-rw------- 1 user2 user2  441 Dec  6 13:39 1765017587.V802I12371eM407537.mail.tahsin.com
-rw------- 1 user2 user2  441 Dec  6 13:43 1765017795.V802I1237daM434313.mail.tahsin.com
-rw------- 1 user2 user2  452 Dec  6 15:58 1765025926.V802I123a4eM355875.mail.tahsin.com
-rw------- 1 user2 user2  440 Dec  6 17:28 1765031289.V802I123b3fM108582.mail.tahsin.com
-rw------- 1 user2 user2  432 Dec  6 17:51 1765032688.V802I123b9dM378132.mail.tahsin.com
-rw------- 1 user2 user2 1294 Dec  6 17:51 1765032691.V802I123b9fM974154.mail.tahsin.com
```

5. Reading the Message and it's Details

```
user2@mail:~$ cat /home/user2/Maildir/new/1765032691.V802I123b9fM974154.mail.tahsin.com
```

The message was:

I am Hasib.

```
Return-Path: <user1@tahsin.com>
X-Original-To: user2@tahsin.com
Delivered-To: user2@tahsin.com
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/simple; d=tahsin.com;
        s=default; t=1765032691;
        bh=Pz9HH3QVxlev+1aHk4v9hOarvk8I44VHsU5mQBYoPo0=;
        h=Date:From:To:Subject:From;
        b=d5v9MDK/3O0jzA8N2QEl4iPOH61xXNusNhFpfGrP5D5vluLarR7zAtJj7W7/K/rgf
         Vp70o7zZ0IF4/hI7aqfft29659lp/V+xNC0f4FCzJa6wXkOqpxXl8eCTVCArIc3r8l
         ZpTsKbrVWFIhaR+ZGRP/SxhLjW6mFcPNk8WKJHsgsXZbvZ86iw2DWOfJ2901kMw1kB
         1sh5L/qMd96Vm2V2IOvGvJuxoMUPQm4Mxd/k17B5evqfbujnWEFQTYF0nEtofauzQA
         RB1UPnt0WQ3afyz3goazPr1fiGEEfEG1MEii/wYgLsgni8dCVNeXplPnIKZL3dad3o
         XXosxQUo7F9FA==
Received: from mail.tahsin.com (localhost [127.0.0.1])
        (using TLSv1.3 with cipher TLS_AES_256_GCM_SHA384 (256/256 bits)
         key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256)
        (No client certificate requested)
        by mail.hasib.com (Postfix) with ESMTPSA id D7CB4123B9E
        for <user2@tahsin.com>; Sat,  6 Dec 2025 17:51:31 +0300 (MSK)
MIME-Version: 1.0
Date: Sat, 06 Dec 2025 17:51:31 +0300
From: user1@tahsin.com
To: user2@tahsin.com
Subject: Who I am
Message-ID: <088a5802792f99688ebd0e0e07713466@tahsin.com>
X-Sender: user1@tahsin.com
Content-Type: text/plain; charset=US-ASCII;
 format=flowed
Content-Transfer-Encoding: 7bit

I am Hasib
user2@mail:~$
```

Check SPF Record (DNS TXT)

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt tahsin.com 127.0.0.1
```

Output:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

tahsin.com      text = "v=spf1 mx ip4:192.168.218.134 -all"
```

SPF = PASS

Because the mail server IP = matches the IP inside the SPF record.

Check DKIM Record

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt mail._domainkey.tahsin.com 127.0.0.1
```

Output:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt mail._domainkey.tahsin.com 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

mail._domainkey.tahsin.com      text = "v=DKIM1; h=sha256; k=rsa; " "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAynoZq
MUFaxOY3v8AFYFcJLdO92k3tvWx5IGoLRDdkkTmcfQNe2yjE0TujByLRb/NFLiUhtM2JBovTOJHbOQvq543cmxvPgUcUgSXtFhGXVwDVNNfsonca7GVgwBWI
okxGMHxJhdEH7871bV1azqs+Y1Wh9tG3zYIvAqV4NLbsdVw7YjRCqkWRV60VgsoyNmsSVPJ1L8iKmlG39" "ToFh/TtMrNVe2Mgxp6dZdWo35fOhI11YPf5R
fGcvaRi8epPb4tEJmh5+QTHhf1+Ljpy5YbV+5TqerXhbbeHaE4DpiNjs2JM8IPnU4nOncLQtZHXz4viNwuCagg3a46VutweqCZHQIDAQAB"
```

DKIM = PASS

Because the DKIM public key is being resolved from DNS.

Check DMARC Record

Command:

```
ahmad@mail:/var/www/roundcube$ nslookup -type=txt _dmarc.tahsin.com 127.0.0.1
```

Output:

```
Server:         127.0.0.1
Address:        127.0.0.1#53

_dmarc.tahsin.com       text = "v=DMARC1; p=none; rua=mailto:admin@tahsin.com"
```

DMARC = PASS

6. Security Analysis

One of the most important parts of this lab was to secure the entire email server and provide strong defense against various attacks, unauthorized access, spam, and email spoofing. Since a mail server is one of the most targeted services on a network, it is essential to configure its security properly. This lab followed modern best practices in everything from TLS/SSL encryption to SPF, DKIM, DMARC authentication, firewall-based port control, and server hardening.

First, TLS/SSL encryption was implemented, which serves as the foundation of email server security. By using TLS in SMTP, IMAP, and POP3 protocols, email data is sent through an encrypted tunnel instead of traveling over the network in plaintext. STARTTLS support was enabled in Postfix and SSL was made mandatory in Dovecot, so that encrypted sessions are used on both IMAPS (993) and POP3S (995) ports. Even though a self-signed certificate is used, the server's encryption strength is quite high due to the use of TLSv1.3, RSA-2048 keys and AES-256 GCM cipher. Testing the TLS handshake with openssl s_client shows that cipher negotiation, certificate verification and session initialization were all completed successfully.

The authentication subsystem uses three modern technologies - SPF, DKIM and DMARC. SPF (Sender Policy Framework) ensures that only authorized servers - in this case mail.tahsin.com - can send emails on behalf of the IP domain. It plays an important role in preventing spam and address spoofing. DKIM (DomainKeys Identified Mail) ensures the authenticity of outgoing emails by cryptographic signatures. In the lab, OpenDKIM was installed, a private key was generated with a default selector, and a DKIM public TXT record was added to Bind9 DNS. The log shows "DKIM-Signature added", which proves that the server is successfully applying DKIM signatures to all outgoing messages. DMARC (Domain-based Message Authentication, Reporting & Conformance) manages policies based on SPF and DKIM results. The DMARC record used in the lab was p=none, which generates authentication reports even if the email is not blocked and helps the domain owner monitor the SPF/DKIM health. SPF, DKIM, DMARC-All three DNS records were resolved correctly from the Bind9 server, resulting in an authentication PASS. Port security was another important component of this lab. In the case of a mail server, only the necessary ports should be open - SMTP (25), Submission (587), IMAPS (993), POP3S (995) and the specific port for IMAP/POP3. Using the ss -tlnp command, it was seen that Postfix and Dovecot were listening properly on the necessary ports and the unnecessary SMTPS (465) was disabled, which is optional according to modern standards. Only the necessary ports were allowed through the firewall using UFW, which protects the server from unnecessary inbound traffic. As a result, effective security was established against brute-force attacks, unauthorized access attempts, and port scanning.

In addition, several important security best practices were followed throughout the lab. Anonymous authentication was disabled in Postfix and Dovecot, secure resolving was ensured by adding OpenDNS or Google DNS fallback to the DNS server, systemd-resolved was disabled and Bind9 was established as authoritative DNS, Maildir permissions were properly controlled via vmail UID/GID, and the Roundcube installer was removed after configuration for security reasons. Even the DKIM private key permissions were made accessible only to opendkim using chmod 600, which is important to prevent server-side key leakage or privilege escalation. Logging and troubleshooting were also considered as part of security. Since /var/log/mail.log is not available in Ubuntu 24.04, Postfix logs are being recorded in /var/log/syslog, and mail processing was monitored in real-time using tail -f. From here, SPF validation, DKIM signing, SMTP queueing, LMTP delivery - everything has been analyzed properly. Structured log analysis is very important in identifying security problems, because the log is the first warning signal in case of brute-force or relay abuse on the mail server.

In this lab, the security of the mail server has been ensured at three levels - transport layer encryption (TLS), identity verification (SPF, DKIM, DMARC) and system-level hardening (Firewall, permission control, service isolation, secure DNS, and authentication restrictions). Modern security guidelines have been followed at each step, which has made it possible to create a secure, reliable and industry-standard email server.

7. Discussion

While completing this lab, I faced several practical challenges that deepened my experience of real server administration. The most notable challenge was the DNS server configuration and the conflict between Bind9 and systemd-resolved. In modern Ubuntu versions, resolv.conf was managed by systemd-resolved, so Bind9 could not run on the same port, which resulted in mail being stuck in the Postfix queue due to external MX record resolution. Using journalctl to detect overlapping bindings between dnsmasq and Bind9, and then disabling systemd-resolved and adding the correct nameservers improved my troubleshooting skills. Another challenge was to properly set up SASL authentication and Postfix-Dovecot socket communication, as even a small mistake in the entire chain of authenticating user to mapping to mailbox storage path could cause an authentication failure in the logs. The Roundcube configuration also required ensuring the correct database credentials, IMAP/SMTP hostname, and PHP-IMAP extension, otherwise the login would fail.

Email security protocols-SPF, DKIM, and DMARC-work as a unified system, although each plays a different role. SPF essentially specifies which IP addresses can send email on behalf of a domain. DKIM provides a cryptographic signature to outgoing email, so that the recipient server can use the public key from the DNS to verify that the email has not been altered in transit. DMARC analyzes the results of SPF and DKIM and applies policies, such as accept, quarantine, or reject. When used together, these three provide strong protection against domain spoofing, phishing, and forgery. The fact that these authentication layers PASSed from the local Bind9 DNS server in the lab proves that DNS plays a crucial role in real-world email security. The authentication trio works together to build trust in the email ecosystem, especially when it comes to verifying the legitimacy of sent emails. This lab clearly demonstrates how the various mail server components-Postfix, Dovecot, Bind9, OpenDKIM, and Roundcube-work together. The SMTP server Postfix handles email sending and queuing, while Dovecot handles mailbox storage, IMAP/POP3 access, and authentication. The Bind9 DNS server provides identity-based mapping, specifying which servers can send and receive emails. OpenDKIM cryptographic identity verification, and DMARC makes decisions based on that authentication data. Roundcube, on the other hand, provides a graphical interface that connects to the user's mail server via IMAP/SMTP. These components, although separate, are interdependent in a complete email system, so that despite the modular design, all parts work as a single ecosystem. In the real world, email server configuration and authentication technologies are widely used. Corporate organizations build their own internal mail servers to ensure data confidentiality, policy control, and central logging. SPF, DKIM, and DMARC are used to prevent spoofing, impersonation, and phishing attacks-the biggest email-based cyber threats today. Roundcube webmail is often used as a lightweight and open-source alternative, especially in small organizations or educational institutions. The techniques learned in the lab-queue troubleshooting, DNS misconfiguration fix, TLS handshake analysis-are fully applicable to real-world server administration tasks. Even cloud-based email systems, hybrid email environments, DevOps-driven mail automation-all require tools like Postfix and Dovecot.

8. Conclusion

This lab provides a comprehensive and practical understanding of building, configuring, implementing security, and integrating various components into a complete email server. From system preparation to Postfix SMTP setup, Dovecot IMAP/POP3 configuration, Bind9 DNS management, TLS/SSL encryption, SPF/DKIM/DMARC authentication, and Roundcube webmail integration, the entire process provides a hands-on experience of building a production-grade email infrastructure. By completing each step, students gain a deep understanding of the email system workflow-mail sending, queue management, cryptographic signing, DNS resolution, mailbox storage, and user access. From a security reflection perspective, this lab demonstrates that a multi-layered approach is crucial to securing an email server. TLS/SSL encryption ensures data security on the network, SPF determines the legitimate sender of the domain, DKIM cryptographic technology ensures message integrity, and DMARC policy provides highly effective defense against spoofing and phishing. Using permissions, authentication restrictions, firewall rules and log monitoring in each server configuration has shown that a properly configured mail server can be not only functional, but also managed securely. Experience in resolving challenges such as systemd-resolved and Bind9 conflicts, DNS recursion problems or SASL authentication mismatches has introduced us to the real problems of server management.

There are some possibilities for further improvements that could make this system even more robust and modern in the future. For example, adding automated SSL certificate provisioning using a CA like Let's Encrypt will further increase the security and trust level of the server. Upgrading the DMARC policy from “none” to “quarantine” or “reject” will make spoofing protection more effective. Using additional mail filtering tools like Fail2ban or spamassassin will further strengthen control over brute-force attacks and unsolicited mail traffic. Two-factor authentication (2FA) integration with Roundcube and rate limiting or outbound spam protection with Postfix will ensure better security in production environments. Overall, this lab project was not just a technical exercise, but rather an effective learning experience to develop a deep understanding of the underlying workings and security of modern email communication systems. This lab provided a complete explanation of the ecosystem that each component of an email server-SMTP, IMAP, DNS, TLS, authentication-combines together to create. This knowledge can be further extended to larger scale or cloud-based infrastructures in the future, which will help build robust, secure, and scalable mail systems.

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